

A BRIEF INTRODUCTION TO THE MODELING AND SIMULATION OF MINERAL
PROCESSING PLANTS

R P KING
Department of Metallurgical Engineering
University of Utah

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CHAPTER 1

SIMULATION OF ORE DRESSING PLANTS - THE BASIC PRINCIPLES

1.1 What is Simulation?

In recent years many references appear in the literature to computer simulation as a technique for design and analysis of mineral processing operations. It has become a fashionable field of research and a great deal of good work has been done to make simulation into a viable and practical tool. Simulation is any procedure that can be used to model a process without actually running the process itself. There are many ways in which a simulation can be achieved but by far the most effective is by digital computer. The digital computer is programmed to mimic the behavior of the actual plant and can provide a description of what the plant will do and how it will perform under a variety of circumstances. It turns out that this is a useful thing to do because the computer can expose many aspects of plant performance without the inconvenience of engineering and operating the plant itself under experimental conditions. It is surprising how effective a good computer simulation can be in providing information about the operation of a mineral processing plant.

Simulation of a complex engineering system is only possible once a detailed understanding of each component of the system has been achieved and simulation provides the engineer with a tool for the prediction of system behavior even if the system does not exist in reality. However the simulator predictions can only be as good as the basic understanding of the component parts. The computer is an essential component of simulation for two reasons: in most systems of interest to the mineral processing engineer, the individual unit operations are so complex that they can only be usefully described in mathematical terms if these can be translated into computer code; in addition the systems of interest reveal complex interactions and interconnections among the individual units. In many cases these complex interactions cannot be described adequately by purely mathematical methods and the ability of the computer to transfer information from one program model to another is exploited effectively to simulate the transfer of actual material, information or energy in a real system. Purely mathematical formations of complex systems are really effective only when the systems are linear in the mathematical sense. Then the full power of linear and matrix algebra can be brought to bear on the problem. Regrettably most systems of real interest are highly non-linear and it is necessary to use the heuristic capabilities of the computer to take the place of purely mathematical descriptions.

The relationship between the model description and the real engineering system will be stressed several times during these lectures. This is vital and is the fundamental principle involved. A computer simulation is an abstract representation of reality constructed in computer code. To be useful it must represent the appropriate aspects of the real situation in such a way that useful

information can be gained. Models for the unit operations are synthesized from mathematical models of the component parts. We will be considering this in detail later in this course.

Simulation techniques have developed tremendously over the years and now it is possible to simulate the performance of a wide variety of systems of engineering interest. We will be interested specifically in the simulation of ore dressing plants. The simulator that we will be using has been developed so that it can provide a realistic description of the operation of any ore dressing plant. It is based on the best simulation techniques currently available and uses efficient numerical analysis procedures to ensure that the computations are done accurately and reliably.

1.2 What is an Ore Dressing Plant Simulator?

An ore dressing plant simulator is a set of computer programs that will give a detailed numerical description of the operation of an ore dressing plant. The simulator must be provided with an accurate description of the ore that is to be processed, a description of the flowsheet that defines the process and an accurate description of the operating behavior of each unit operation that is included in the flowsheet. The simulator uses these ingredients to provide a description of the operating plant. The detailed description of the ore will include information on its physical and mineralogical characteristics and a method to provide this information is described later. The flowsheet is the familiar graphical representation of the location of the unit operations in the plant together with the network of pipes and conveyors that transmit material between the units. The description of the operating behavior of the unit operations is what is known as modeling and it is necessary to accurately model the behavior of each unit operation before the entire plant can be simulated. In a sense the simulator links together the modeled behavior of each of the unit operations and synthesizes the overall performance.

Four fundamental concepts underlie the construction of an ore dressing plant simulator.

- * Ore dressing plants are collections of unit operations connected by process flow streams that transmit material from one unit to the next. The flow of materials is directed by the flowsheet structure.
- * Each unit operation processes its own feed materials and will separate it or transform it in accordance with the specific objective of the unit.
- * The behavior of the plant as a whole depends on the operating characteristics of each of the unit operations as well as on the nature of the material that is processed in the plant.

- * A simulator reduces the actual plant operations, as defined by the flowsheet structure and the behavior of the units, to a sequence of logical mathematical functions. The simulator can then mimic the real plant performance.

We will examine each of these concepts and discuss how they lead to the construction of a viable and reliable simulator. The basic concepts are independent of the precise nature of any particular plant that must be simulated and they lead to the development of simulation software that can be used for all possible plant configurations. The availability of such general purpose software makes computer simulation a useful practical tool in every day engineering. It is a difficult task to write the necessary computer code to simulate a complex ore dressing plant. Most engineers have neither the time, inclination nor skill to do so and it would not be cost effective to write the code for each application. The cost in man hours to generate the code and debug it so that it can run reliably would be enormous. Computerization of any complex engineering systems is a highly specialized business and this is true also in mineral processing and such activities should be attempted only by specialists. Several general-purpose simulators for ore dressing plants are now available and of these MODSIM offers the greatest versatility to the user to modify and adapt the models of the unit operations that are used by the simulator. The standard models that are provided in the package are based on the latest concepts from the modern mineral processing literature. Several of the models that are incorporated as standard have been developed by the author and are not available in any of the commercial simulators. MODSIM is particularly strong in the modeling of mineral liberation phenomena.

In order for a general-purpose simulator to yield useful information on the actual process to be simulated, it must have access to 3 important classes of information. These are defined in general terms as follows:

- The structure of the flowsheet - what unit operations are included and how they are connected.
- The nature of the material to be processed - its mineralogical composition and structure, the size distribution and the amount that must be processed.
- The operating characteristics of each unit in the flowsheet. This requires the full description of the unit operations (the unit models) and a specification of the unit parameters that define the operating characteristics of the individual units.

1.3 What can a simulator be used for?

A good simulator is a useful tool to the process plant engineer. Essentially the simulator can demonstrate what a plant will do under any particular operating conditions. It can do so cheaply and without any real risk to the production rate of an operating plant or it can do so before a plant has been built and it does so in the engineer's office.

A. Design studies:

At the design stage a good simulator can be used to

- help the design engineer to find the best flowsheet
- ensure that design specifications will be met under all required operating conditions
- choose the most suitable units
- size the units correctly and so eliminate wasteful over-design and avoid the catastrophe of under design
- optimize the plant operation by achieving best economic combinations of grade and recovery
- identify potential production bottle necks
- provide comparative assessment of competing manufacturers' equipment
- define the performance guarantees that should be met by suppliers
- find out what will happen if performance guarantees are not met.

B. Operating plant performance:

A good simulator can help the plant manager to

- get the optimum performance from his plant
- tune his plant to suit variations in feed quality
- find plant bottlenecks
- investigate performance changes by asking "what if" questions
- identify unit operations that are not properly understood

- make better use of those that are.

But remember that a simulator can be effective only if it gives a reliable and valid description of plant operations.

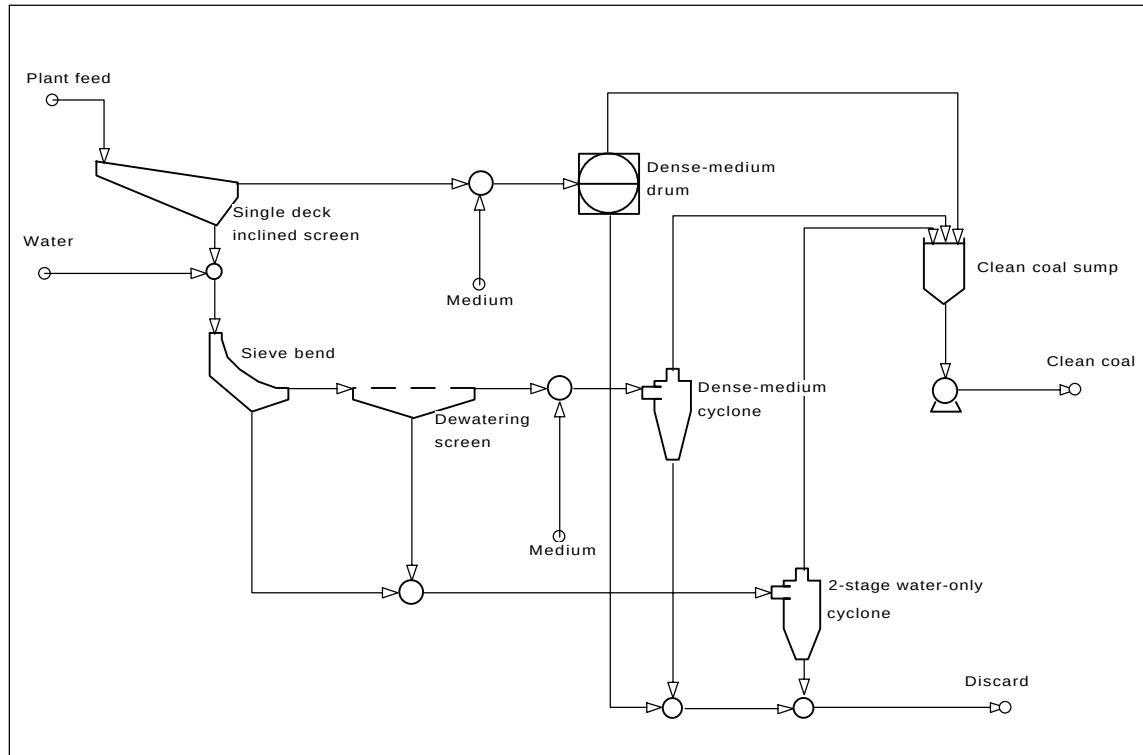


Figure 1.1 A typical plant flowsheet with three separate concentration units; a dense-medium drum, a dense-medium cyclone and a water-only cyclone.

A typical flowsheet of a coal washing plant is shown in Figure 1.1. The overall performance of a plant such as this depends critically on the densities at which each of the concentrating units are set to operate. There is no simple way to determine the optimal combination of settings but they can be established using a straight forward application of simulation techniques. Studies on real plants have shown that increases in the yield of product of 4-5% can be achieved without any sacrifice of grade when the plant is set up optimally. Gains of this magnitude can make significant improvements to the profitability of a large production plant. The only alternative to simulation is a costly and time consuming set of plant trials that must be repeated whenever one or more significant changes occur in the feed material. Plant trials always require periods of non optimal operation with an inevitable loss of production.

1.4 The Ore Model

Ore dressing is the process engineering technology that must necessarily come between the mining of mineralogical raw materials and the subsequent extraction processes that recover the useful metals. Ore dressing is directed primarily at the separation of individual minerals or groups of minerals from among the whole range of minerals that make up an ore body. Minerals are, for the most part, crystalline inorganic chemical species and the variety of crystal types is enormously large. Minerals are the raw materials from which are recovered individual elements that are chemically locked within the inorganic crystalline material.

The extraction of these elements by pyro- or hydro metallurgical processing routes is comparatively costly and the concentrating action of ore dressing processes can have a significant effect on the cost of the extractive metallurgical processes. Because no chemical transformations are involved in ore dressing, the dominant processes involved are those which transform the physical nature of material and separate solid material using a number of physical attributes of the solid materials to achieve the separation.

The crystalline nature of most mineral ores that occur in nature dictate that the minerals are present in more or less random admixture on a size scale that ranges from a few tenths of a micron up to a few centimeters. The physical separation of one mineral species from another can be achieved by the reduction of the solid material to the particulate state followed by the separation of particles by one or other physical means. The reduction to the particulate state is essential and the technology of any concentration processes is intimately linked to the nature of the particulate material that is produced by comminution of the parent ore. Indeed the transformation of the size of the material by comminution processes is the dominating transformation process that will be of concern to us.

The separation processes that are used to separate valuable mineralogical material from its associated waste or gangue are amazingly varied relying on a wide variety of physical properties of the minerals to effect a separation. The physical properties that are most widely used are the specific gravity and the magnetic susceptibility of the solid material. Subtle chemical effects, notably the chemical properties of the mineral surfaces are also used with notable success to achieve the separation.

The emphasis in all of ore dressing technology is on the solid material in the particulate state and it is our ability to separate particles from each other that determine the success or otherwise of the ore dressing operation. The principles involved in all the unit operations of interest are associated with the behavior of individual or groups of particles in the equipment. In the early days of ore dressing technology, (the historical records extend back over some two thousand years), the processes were developed on the basis of an empirical understanding of the behavior of particulate solid material in a variety of different situations. During the past three decades or so, the emphasis has shifted to an

analysis of the particle mechanics in the unit operations that have evolved on the basis of the earlier empirical understanding. The basic principles of all the major unit operations of ore dressing are now comparatively well understood and these basic principles can be used to analyze the particle mechanics involved in each unit operation. The operation of the unit as a whole is determined by the sum total of the behavior of all the solid particles as they are processed.

MODSIM is designed to exploit the modeling techniques that are based on the particle mechanics of the ore dressing unit operations. The models used in MODSIM are based on the particulate nature of the solid material that is processes and, as a result, models of considerable complexity can be devised and used for the description of the unit operations. This ability to accommodation models of considerable complexity gives MODSIM their versatility and power and a thorough understanding of the particulate description of the solid material is required to take advantage of the many advances that have been made in ore dressing modeling techniques during the past three decades.

1.5 The Particulate State

The particulate state is usually defined as describing solid material that is reduced in size to particles that range from a few hundredths of a micron to a few centimeters. For solid material of mineralogical origin this size range implies that 1 kg of material will consist of a large number of particles (ca 1 billion cubes of size 10 μm). Commercial ore dressing plants process many thousands of tons of raw material per day so that the number of particles involved is exceedingly large.

Although the models of ore dressing unit operations are based on the behavior of individual particles in the environment of the process equipment it is quite out of the question to attempt to keep a track of each particle as it passes through the unit operations let alone the complete plant flowsheet with its multiplicity of units and complex flow topology. It is this fact more than any other that has determined the data structures that are required to give a valid description of the particulate solids and still be comprehensive enough to exploit the basic concepts of particle mechanics to provide a description based on sound physical principles.

Because of the large numbers of particles involved it may be tempting to look to chemical processing to provide an analogy for the model structures. This is precluded because of one significant fact: the basic entities for chemical processing are molecules which for a given compound have invariant properties while for ore dressing the basic entities are particles, no two of which can ever be identical. We are therefore faced with the problem of devising a mathematical description that is capable of accommodating the large population of individual particles while still acknowledging the variation of properties from particle to particle. Fortunately a powerful mathematical structure, which borrows

heavily from the mathematical theory of probability is readily available and is forming the basis for all of the modern developments in the modeling of ore dressing operations.

The description of the particulate material is based on the concept of the distribution function which is most familiar as a descriptor for the particle size distribution. The notation is straightforward and is simple to relate to measurements that are made in the laboratory as is shown in the following section.

1.5.1 The particle size distribution

The most obvious characteristic of a particle that is significant from the point of view of its behavior in an ore dressing operation is its size. Size itself is difficult to define precisely but fortunately the method of description to be used does not require precise definition nor precise experimental determination of particle size. Considerable leeway is permissible without compromising the accuracy of the method. The most common measure of particle size is the smallest square wire mesh that will permit the passage of the particle under gravity with prolonged shaking. Such a size is readily measured in the laboratory to sufficient precision for practical purposes by the familiar sieve analysis procedure.

The essential feature of the mathematical description is the particle size distribution function $F(d_p)$ defined as follows:

$F(d_p)$ = mass fraction of particles in the population that will pass through a square mesh sieve having an opening with side = d_p

$P(d_p)$ is an ordinary function of d_p but it does have some important properties that should always be borne in mind. These properties are:

$$\begin{aligned}P(0) &= 0 \\P(\infty) &= 1 \\P(x) &\geq P(y) \text{ whenever } x \geq y\end{aligned}$$

The value of P is measured experimentally at a number of fixed sizes that correspond to the mesh sizes of the set of sieving screens that are available in the laboratory. This data is usually presented in tabular form showing mesh size against fraction smaller than that mesh. Graphical presentations are useful and are often preferred because it is generally easier to assess and compare particle size distributions when the entire function is immediately visible. A variety of different graphical

coordinate systems have become popular with the view to making the distribution function plot as or close to a straight line. The particle size axis, usually the abscissa is plotted on a logarithmic coordinate scale. The ordinate scale works according to whether the distribution function $P(d_p)$ is close to log-log, log-normal or Rosin-Rammler. Specially ruled graph papers are available for this purpose and three typical rulings are given in the appendix. The mesh sizes in the standard sieve sizes vary in geometric progression with each mesh size a constant factor larger than the previous one. The constant factor is usually a fractional power of 2 (very often $\sqrt{2}$). Such a geometric series will plot as a series of equidistant points on a logarithmic scale.

Although the distribution function $P(d_p)$ is perfectly well defined and is amenable to direct measurement in the laboratory, it is not directly useful in most cases for modeling of ore dressing unit operations. For this purpose a derived density function is used. The discrete particle size density function $p_i(d_p)$ is defined as follows:

$$f(d_p) \Delta d_p = F(d_p + \Delta d_p) - F(d_p)$$

= mass fraction of the particle population that has size between d_p and $d_p + \Delta d_p$

Δd_p is the so-called size class width and is usually not constant but successive values form a geometric series. This leads to the idea of a particle class which includes all particles in the population which have properties falling in a narrow size interval or class. If the interval is sufficiently small it is possible to assign a single value to the property that defines the class so that each particle in the class may be assumed to behave as a particle having the class average property. In the case of particle size this representative size is generally taken as the geometric mean except for the two extreme classes which have no geometric mean. It is usual to extend the average sizes as geometric series to the two extreme classes and this is generally satisfactory in practice. MODSIM operates with the individual particle classes and consequently the particle size distributions must be specified in the differential form.

The key to the success of this approach to ore dressing modeling is the use of narrow class interval so that the behavior of all particles in the class can be realistically modeled using the class-average property. This in turn implies that a large number of particle classes must be specified. MODSIM is created specifically to handle large numbers of particle classes efficiently and is thus an ideal vehicle for the simulation of ore dressing flowsheets using accurate particulate models. The extra effort required to make these accurate calculations is removed entirely from the user.

Table 1 Particle size distribution in the rod mill product.

Size class	Size interval microns	Representative size microns	Particle size distribution density %
1	+2360	2780	2.4
2	-2360 +1700	2000	3.1
3	-1700 +1180	1416	4.0
4	-1180 +850	1000	5.0
5	-850 +600	714	6.6
6	-600 +425	505	9.1
7	-425 +300	357	13.1
8	-300 +212	252	16.4
9	-212 +150	178	12.7
10	-150 +106	126	7.6
11	-106 +75	89	4.9
12	-75 +53	63	3.7
13	-53 +38	45	2.8
14	-38 +27	32	1.8
15	-27	22	6.8

A typical specification of the size distribution for the feed to a ball mill circuits is shown in Table 1. This is based on a standard $\sqrt{2}$ series and covers the range from 2,78mm to approximately 30 microns in 15 classes. MODSIM will normally use more size classes for its internal calculations than are used to specify the size distribution of the plant feed streams. The Users Manual should be consulted for details of the methods that are used to specify the size distributions and to select the number of size classes to be used in the calculations.

1.5.2 Mineral Liberation and the Grade Distribution

The mineralogical composition of the particles that are processed in ore dressing operations varies from particle to particle and this is of paramount importance in the operation of ore dressing equipment. The very objective of ore dressing operations is the separation of mineral types to produce concentrates having a greater relative abundance of the desired mineral. The objective of the comminution operations is the physical separation of the minerals by fragmentation.

Unfortunately, except in very favorable cases, the minerals do not separate completely and many particles, no matter how finely the material is ground, will contain a mixture of two or more mineral species. Some particles will indeed be composed entirely of a single mineral and they are then said to be completely liberated. The amount of mineral that is liberated is a very complex function of the crystalline structure and mineralogical texture of the ore and the interaction between these and the fracture pattern that is induced by the comminution operation. Considerable research has been devoted to the liberation phenomenon in recent years and several good models are available to describe the liberation characteristics of an ore. In order to properly allow for incomplete liberation of the mineral species an additional distribution function is defined but before this is done it is necessary to devise a method for representing the mineralogical composition of a particle.

When only two mineral species are involved, say a valuable mineral and a gangue, this is not difficult. It is necessary only to specify the mass or volume fraction of the particle that is composed of mineral. We normally use the symbol g to represent this fraction (the grade of the particle). When more than two mineral species are relevant the situation is somewhat more complicated and g will be a vector having more than one component, each component representing the mass fraction of a single mineral species. The fractions must accordingly sum to unity. To handle this complexity the concept of the distinct particle type is defined. Just as in the case of the particle size, finite classes are defined each of which is characterized by particles of an average mineralogical composition. The number of classes that should be used will depend on the mineralogical complexity and the liberation characteristics of the ore that is to be processed in the simulated process.

One class is usually allocated to each of the pure minerals that are present because it is assumed that

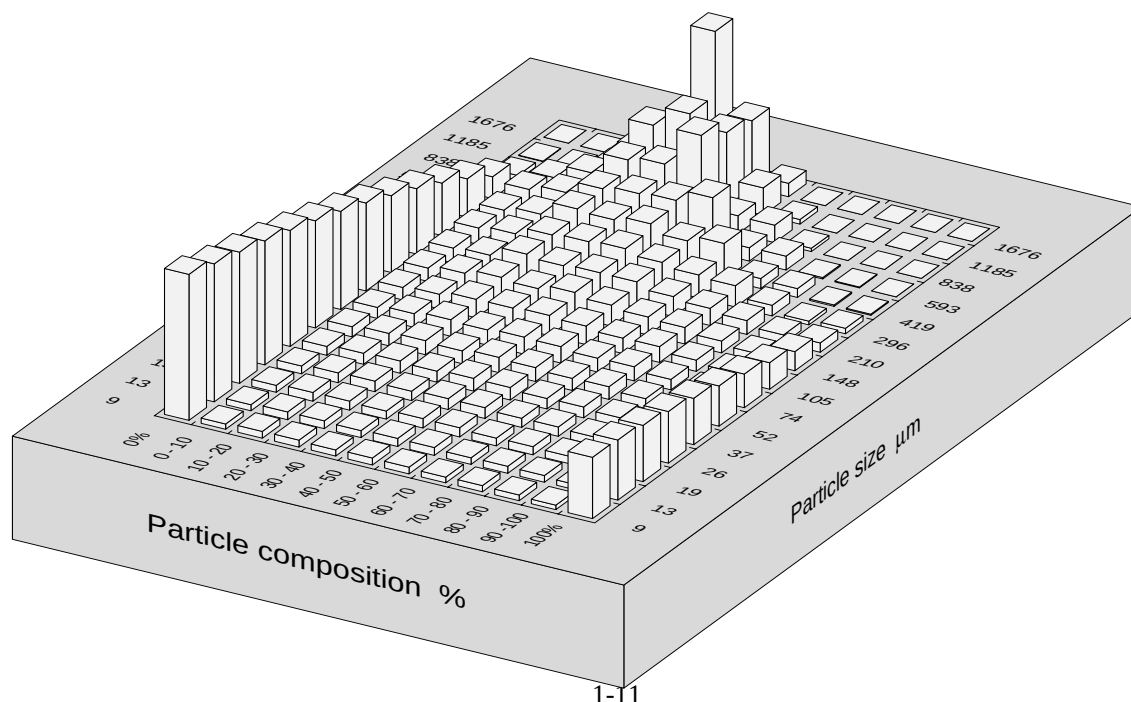


Figure 1.2 A typical representation of the distribution of two-component particles among the grade and size classes.

some of each mineral is perfectly liberated. MODSIM will automatically allocate such perfectly liberated classes but will only allocate particle classes to accommodate incompletely liberated minerals if specifically requested by the user. If such a request is made, the user must define the particle composition that is required to define each particle type. A typical set of data is represented graphically in Figure 1.2, 1.2. Here an ore is classified in 16 size classes and 12 grade classes. The representative sizes of the size classes are shown along the size axis and the composition of range of each grade class is shown along the composition or grade axis. The height of each bar represents the fraction of material in the population that occurs in each combination of grade and size class. Note that in Figure 1.2, 1.2 the data is normalized so that the sum of the fractions in the grade classes will sum to unity for each size class. This method of representation is called a conditional distribution of the particle grade for each size class and it is used to show more clearly how the mineral liberates as the size of the particles decreases.

Table 2 Washability data for coal

Particle Type	Relative Density Interval	Fractional Yield %	Cumulative Yield %	Ash Content %
1	F1.30	18.64	18.64	4.30
2	1.30-1.32	11.15	29.79	5.11
3	1.32-1.34	6.67	36.47	6.22
4	1.34-1.36	7.12	43.58	7.15
5	1.36-1.38	6.87	50.45	9.04
6	1.38-1.40	6.22	56.67	10.80
7	1.40-1.42	5.47	62.14	12.20
8	1.42-1.44	5.04	67.18	13.90
9	1.44-1.46	5.02	72.20	16.10
10	1.46-1.48	4.60	76.80	17.90
11	1.48-1.50	3.77	80.57	21.30
12	S1.50	19.42	100.0	40.60

Another common example of this scheme for the classification of particle types is the familiar washability data for coal. The standard washability analysis separates the coal into a number of narrow relative density classes each one of which can be characterized by the ash content of the coal. The greater the ash content the greater the relative density. A typical washability analysis for a sample of coal is given in Table 2

Table 3 Data from Table 2 specified in a format suitable for use within MODSIM

Particle type	Mineral distribution for this type	Distribution density of particle types	Specific gravity of this type
1	0.9570 0.0430	0.1864	1.10*
2	0.9489 0.0511	0.1115	1.31
3	0.9378 0.0622	0.0667	1.33
4	0.9288 0.0712	0.0712	1.35
5	0.9096 0.0904	0.0687	1.37
6	0.8920 0.1080	0.0622	1.39
7	0.0878 0.1220	0.0547	1.41
8	0.8610 0.1390	0.0504	1.43
9	0.0839 0.1610	0.0502	1.45
10	0.8210 0.1790	0.0460	1.47
11	0.7870 0.2130	0.0377	1.49
12	0.5940 0.4060	0.1942	1.67*

* estimated from a plot of % ash vs l/sp. gravity

The equivalent data specification for use within MODSIM is given in Table 3. Note that no particle classes have been allocated to completely liberated particles because perfectly liberated coal and ash particles are never observed.

For accurate and reliable simulation of coal washing plants it is necessary to specify the washability of the coal in as many size fractions as possible. MODSIM can accommodate such data easily and in fact will give greatly enhanced performance when more complete data are available.

A number of convenient data input formats that are specific to coal are provided in MODSIM. If the proximate analysis is available for each washability fraction in each size class, this data can be input directly and MODSIM will then calculate the proximate analysis of the material in every stream of the plant. This will give a very complete description of the coal washing operation. MODSIM will require as a minimum the ash content for each washability fraction. MODSIM will also keep track of the calorific value and the sulfur content (both pyritic and organic) in each process stream including the products. See Table 4.5 for an example of the summary output from the simulator when the proximate analysis, the calorific value and the sulfur content are known in each washability fraction of the feed material.

No comparable standard data formats are in common use for other mineral systems so that the mineralogical composition must be specified for each particle type. This is the liberation spectrum for

the ore. It is now becoming quite common to see the liberation spectrum for binary ores specified in 12 particle classes - one for each of the liberated minerals at each end of the spectrum and ten classes that describe the mixed particles at intervals of 0.1 in *g*. Ores with more than two mineral components that must be described using more particle type classes and it is only recently that liberation models for multi-component materials have been developed.

The liberation spectrum is of course a strong function of the size of the particle. In general the finer the particle the more likely to find particles that are completely liberated. It is necessary therefore to specify the distribution of material among the particle types for each size class. However, if the plant that is to be simulated includes comminution units, it is often sufficiently accurate to assume that all the feed material is concentrated in the particle type class that includes the average mineral composition. The liberation of the minerals will be generated by MODSIM's liberation model. MODSIM cannot accommodate variations in the mineral distribution for the grade classes for the various particle sizes. In general that would not be necessary even for the most detailed of unit models.

1.6 Physical Properties of the Particles

All ore dressing operations rely on one or another physical property to effect the desired concentrating action in the unit. Since the objective of ore dressing is the separation of the valuable minerals from the others, it is necessary that the physical properties vary from particle to particle. The variation of physical properties can very conveniently be accounted for by specifying the value of the physical property for each particle type. Each particle in the type class may then be considered to have the value of the physical property assigned to that type class.

The most important of the physical properties that is used in ore dressing is the particle specific gravity. This property forms the basis of all the gravity, dense medium and centrifugal separation processes and, after flotation, these processes account for the largest tonnage of material processed. The specific gravity of a particle is determined by its mineralogical composition. Thus the specific gravity of a composite particle is a weighted sum of the specific gravities of the constituent minerals with the weighting coefficients being the volumetric fractions of the mineral components. MODSIM therefore will compute the specific gravities of each particle type once the distribution of minerals has been defined for each particle type and the specific gravities of the individual minerals have been specified. This is the natural method of specification and relieves the user of the burden of calculating a specific gravity for each particle type. However in some situations it is not possible to specify the specific gravity of the pure mineral species. This is true for example with coal since the carbonaceous material in coal is made up of a mixture of macerals and definite specific gravities cannot be assigned

to the individual mineralogical components. Under these circumstances it is necessary to determine the specific gravity of each particle type by direct observations. This is precisely the situation with coal when the washability function is determined because the mean specific gravity of each washed fraction is immediately known from the specific gravities of the liquids at each stage of the washability analysis. This is illustrated in Table 3. MODSIM will request the user to specify whether specific gravities of individual minerals will be specified or whether the specific gravities of the particle types will be specified.

Other physical properties may also be specified and the user can specify as many as are required to suit the models that are ultimately to be used in the simulation.

Within MODSIM the physical properties are identified by number and are available to the model subroutine through the array PPROP. The disposition of physical properties in this array is indexed according to array INDPP as described in section 4.7 of the users manual. This gives the user access to the physical properties for use within his model subroutines.

1.7 Distribution of Flotation Rate Constants

The distributed rate constant model for flotation has become so widely accepted that special provision has been made to specify a distribution of rate constants as primary data in MODSIM. Each particle type can have associated with it a distribution of flotation rate constants. The user is required to specify the total number of rate constants that will cover all the particle types. The distribution of particles over these flotation rate constants must be specified for each particle type. The distribution over every rate constant must be specified but zero values are permitted.

As an example consider the situation in which two minerals are to be separated by flotation. Assume that the mineralogy and liberation characteristics can be adequately accounted for by classification into three particle types: liberated mineral, liberated gangue, and a middling. Many mineral systems have a flotation behavior that can be characterized by classifying each particle type into a floatable and non floatable component. It is in fact remarkable how accurately many flotation systems can be modeled on this simple basis. Assume that this description is adequate for the present example and then four flotation rate constants will be required and the distribution of particles over these rate constants for each of the particle types may be as shown in Table 4.

Table 4 Example of Distribution of Flotation rate Constants.

Particle Type	Distribution of particles over rate constants			
	k_1	k_2	k_3	k_4
1. Liberated mineral	0.8	0.0	0.0	0.2
2. Middlings	0.0	0.7	0.0	0.3
3. Liberated gangue	0.0	0.0	0.5	0.5

Here $k_1 \geq k_2 \geq k_3 \geq k_4 = 0$

A further facility is provided by MODSIM regarding the specification of flotation rate constants. In general the flotation rate constants for a particular particle type will vary from flotation bank to flotation bank. This could be due to interstage addition of reagents or to changes in the chemical environment as pulp moves as flotation concentrates to cleaner stages. Another factor that contributes to variation of flotation rate constant is the change in the nature of the froth particularly in cleaner plant simulation, the user can specify whether he wishes to specify special value of the flotation rate constants at every separate bank of flotation cells. The distribution of particles of a particular type over the rate constant classes may not however be changed. In other words any particle that is in the slow floating class remains in that class. It is only the value of the rate constant for that class that changes.

1.8 Plant Feeds and water Addition

MODSIM can accommodate multiple feeds to the plant that is being simulated. The characterization of the feed material for each feed must be done according to the methods described in the previous sections. Material in all of the feeds must have identical class structures but the distribution of particles over the classes can vary from feed stream to feed stream. For example if the plant takes a feed of coarse material and a feed of fine material, the particle size distribution will be very different for the two feeds. However both feeds must have their size distributions specified against the same size class intervals. Obviously the coarse feed will have none or very little material in the fine size classes and the fine feed will have nothing in the coarse size classes.

The tonnage and water content of each feed must be specified. Dry feed is permissible. Water feeds may be added although these should be added to mixers or sumps only, never directly to a unit. This is not an important restriction since it is always possible to precede any unit with a mixer if water is to be added to the unit. Two options are provided for the specification of water addition rates: the absolute addition rate may be specified or alternatively the user can specify the percentage solid that is required in the stream that leaves the mixer or sump to which the water is added. In the latter case, MODSIM will continuously adjust the water addition rate to match the calculated total solid rate in the sump discharge to maintain the requested percentage solids. This device is very useful for the simulation of control actions that are incorporated to maintain a specified solids content in the slurry at any point in the flowsheet.

1.9 Models for Ore Dressing Operations

The modular design of MODSIM has been developed specifically to give the user complete freedom in the choice of models for the unit operations. The only restriction on the models is the basic structure that requires that a unit model should be capable of receiving the details of the unit feed from the simulator and producing the appropriate product streams. This, of course, is the natural function of a unit model since it mirrors the actual behavior of the unit which transforms a feed material into the appropriate product streams. The details of the feed material provided by the simulator to the model subroutine is in accordance with the particulate model of the solid phase that has been described above. The subroutine is supplied with the mass flowrate to the unit in every particle class that has been defined. The parameters for the appropriate unit model as well as the vector of physical properties are also available to the model subroutine. The water rate in the feed is also supplied. This information gives the user almost unlimited scope to include models as simple or as complex as is desired to model the unit operations. Details of the subroutine structure are given in the User Manual.

CHAPTER 2

SIMULATION OF ORE DRESSING PLANTS - THE SIMULATOR STRUCTURE

In this section the structure of the MODSIM simulator is described. It is necessary to have an understanding of the basic structure so that the user can make effective use of the software package and exercise the necessary control over its operation to ensure that the simulator generates a solution to the problem that the user specifies. MODSIM was designed to be used by the entire spectrum of users from those who will not do any programming at all to those who will want to enhance and develop the system for specific purposes. MODSIM is flexible and has a modular structure. The flexibility is required to ensure that the user can solve a wide variety of different problems and the modular structure is designed to allow the system to accommodate an unlimited and ever-increasing number of modules that describe a variety of unit operations.

A plant flowsheet consists of a number of unit operations connected together by process flow streams. Material enters the plant and passes successively through the process units in accordance with the flowsheet structure, sometimes recycling on itself until finally it leaves the plant in one or other product stream. The material in a flow stream is processed by the unit in order to change the nature of the material or to separate it into its constituent parts. It is fortunate that the number of different types of unit operation encountered in ore dressing plants is comparatively limited and these can be classified into three different generic types.

2.1 The Ore Dressing Unit Operations

There are only three basic types of ore dressing unit operations. separation units that separate solid particles without changing their characteristics e.g. flotation, gravity separation, etc.; transformation units which alter the characteristics of the solid particles e.g. crushing and grinding operations; and units that separate solid and liquid constituents. It is remarkable that the entire repertoire of ore dressing unit operations can be classified by such a simple scheme and this simplicity is exploited effectively in the simulators.

A further characteristic of the unit models is the possible flow stream connections that are possible. Separation units will always produce at least two product streams, a concentrate and a tailing, an underflow and an overflow, etc. Sometimes a single separation unit will produce a middling product as well. On the other hand, transformation units only produce a single product stream. Water separation units always produce two product streams, one of which is always predominantly water.

A convention is used in MODSIM which only allows a single feed stream to enter any unit. In cases where the flowsheet requires a unit to accept more than one stream, these are considered to be mixed before the unit and the combined stream is the unit feed. This does not cause any loss in generality and provides an important advantage for users who wish to develop and insert their own models into the simulator.

The simple structure just described defines the requirements that any mathematical model must meet if it is to be suitable to describe the operation of a unit operation within MODSIM. The mathematical model of a transformation unit must provide a description of the product stream from the details of the feed stream and the operating conditions of the unit. For example, the model of a grinding mill must calculate the particle size distribution of the product from the particle size distribution of the feed material and the breakage characteristics of the solid material that is processed. This latter will depend on the type of material as well as on the type of grinding mill (ball, rod, pebble, autogenous, etc.) and on the milling environment in the mill (ball charge, pulp filling, mill diameter, pulp viscosity, etc.).

The mathematical model of a separation unit must calculate the composition and characteristics of the various product streams that are produced. For example, the model of a cyclone should calculate the size distributions in both the underflow and overflow using the particle size distribution, flowrate and density of the feed as well as details of the cyclone geometry. Likewise the model of a flotation cell must calculate the mineralogical composition of the concentrate and tailings streams from the composition, size distribution and flowrate of the feed to the cell and from the volume, aeration rate and froth height in the cell.

The calculated product stream characteristics are then passed on by the simulator as feeds to the next units in the flowsheets. The models of these units then produce corresponding product streams that are passed on by the simulator. Eventually the material is traced completely through the plant and all the process streams are then defined. In the event that a stream is physically recycled, an interactive procedure is required because the feed streams to some units will not be known until the product streams from some subsequent units have been calculated. MODSIM will handle all such recycle loops iteratively without any interaction from the user. Efficient convergence procedures are provided to ensure that computation time is minimized for the user.

The sequential calculation procedure described above leads to an obvious division of activities within the simulator - the calculation of the performance of each unit and the transmission of the details of the various flow streams from unit to unit. The latter is taken care of by the simulator executive and the former by individual unit models that are inserted as modules into the system and which can be

called by the executive as required. This approach to simulation is known as the sequential modular approach and has proved to be superior to any other method because of its unique advantages.

2.2 Unit Models

The heart of a good simulator is the mathematical model of the unit operation. The simulator can predict reliably and accurately only if the unit models describe the unit operations accurately. Great versatility is required to do this because of the very wide variety of models that have been developed. The best models are developed from a fundamental understanding of the physico-chemical processes that underlie each unit operation and they acknowledge specifically the intricate details of the particulate nature of the material to be processed.

The field of mathematical modeling research is very wide and many significant advances have been made in the techniques that can be used to construct useful models. The literature contains many excellent papers and reviews covering this field and most of the best models available in the literature for virtually all types of ore-dressing unit operations have been incorporated into MODSIM. Although many simulation problems can be solved using comparatively simple models for the unit operations, and these are not difficult to develop and program, it is recommended that the development of more detailed and mathematical models for the unit operations be left to specialists. The simulators make these models available to the engineer in the form of reliable and effective computer code which is easily used effectively.

2.3 Why a Modular Structure?

The sequential modular structure has several unique advantages. In particular it allows models to be inserted as modules into the simulator so that new models can be added whenever required.

The advantages of the sequential modular structure can be summarized as:

- * Versatility to describe any ore dressing plant
- * Different preferences for unit operation models by different users can be easily accommodated.
- * A variety of models is available for most unit operations. It is usually desirable to investigate and compare several of these during simulation studies.

- * It is most convenient to construct and test unit models as stand-alone modules.
- * Models must always be verified and calibrated against operating data. This is done most effectively by comparison with data from single-unit tests.
- * It is cost effective to make use of existing models with minor modifications and extensions to describe new units or operating regimes.
- * Allows for easy exchange of models with other users.

2.4 The Numerical Analysis of Simulation

The models of the various unit operations that have been described in previous chapters have one thing in common - they are all structured in such a way as to make it possible to calculate the nature of each of the product streams from a knowledge of the characteristics of the particulate material that makes up the feed stream to the unit.

The units that have been modeled can be classified into three categories and it is useful to give a general description of each category.

2.4.1 Transformation Units

These units are exemplified particularly by the comminution operations - crushing and grinding. These unit operations are designed specifically to transform the size distribution of the feed stream to a product containing finer material. This is shown schematically in Figure 2.1.

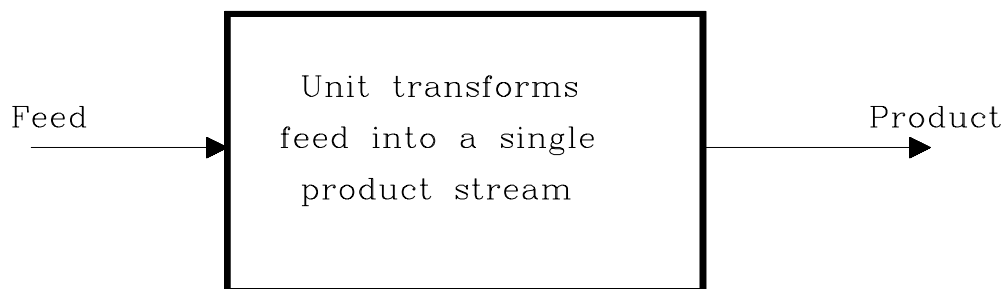


Figure 2.1 *Transformation units produce only a single output.*

The behavior of the unit as a whole can be described by

$$\mathbf{W}_p = \mathbf{T}\mathbf{W}_f \quad (2.1)$$

where $\mathbf{W}$ is a vector of mass flowrates in the particle classes is a particular stream and $\mathbf{T}$ is a transformation matrix that defines how the size distribution of the feed is transformed to that in the product. Subscripts p and f refer to product and feed respectively.

2.4.2 Separation Units

The vast majority of units in an ore-dressing plant are designed to separate particles on the basis of some physical parameter. The efficiency of transfer of particles of a particular type to the concentrate,

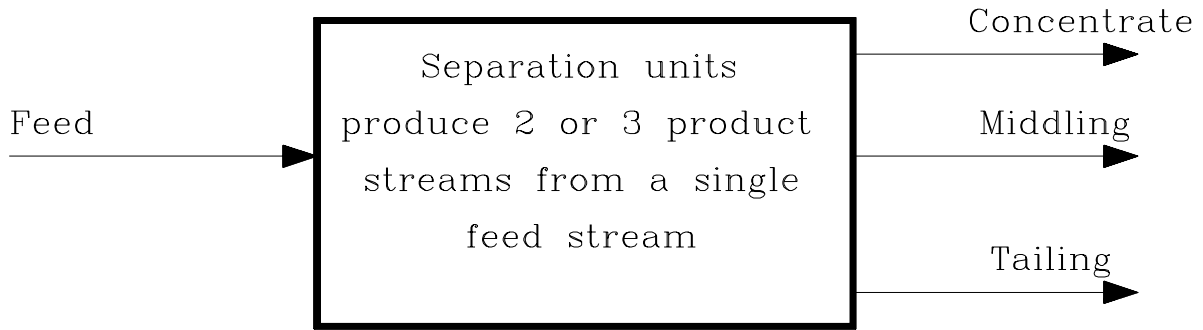


Figure 2.2 *Separation units produce 2 or 3 product streams.*

Middlings, and tailings can be expressed in matrix -vector notation in terms of three diagonal recovery matrices, one for each product stream, $\mathbf{E}_C$ representing recovery to the concentrate, $\mathbf{E}_T$ the recovery to the tailings and $\mathbf{E}_M$ to the middling. The i th diagonal elements of these recovery matrices represent the recovery of particle of type i to the relevant product.

$$\begin{aligned} \mathbf{W}_C &= \mathbf{E}_C \mathbf{W}_f \\ \mathbf{W}_T &= \mathbf{E}_T \mathbf{W}_f \\ \mathbf{W}_M &= \mathbf{E}_M \mathbf{W}_f \end{aligned} \quad (2.2)$$

The concentration of a particular metal component in a stream can be calculated from the elements of the mass flowrate vector. For example, the assay of Cu in the concentrate stream is given by

$$a = \frac{\sum a_i W_i}{\sum W_i} \quad (2.3)$$

where a_i is the assay of Cu in particles of type I .

2.5 The Sequential Method of Simulation

Any ore-dressing plant can be synthesized as a complex interconnection of the basic unit types as shown in Figure 2.3. Products from one unit becomes the feed to another unit or perhaps a final

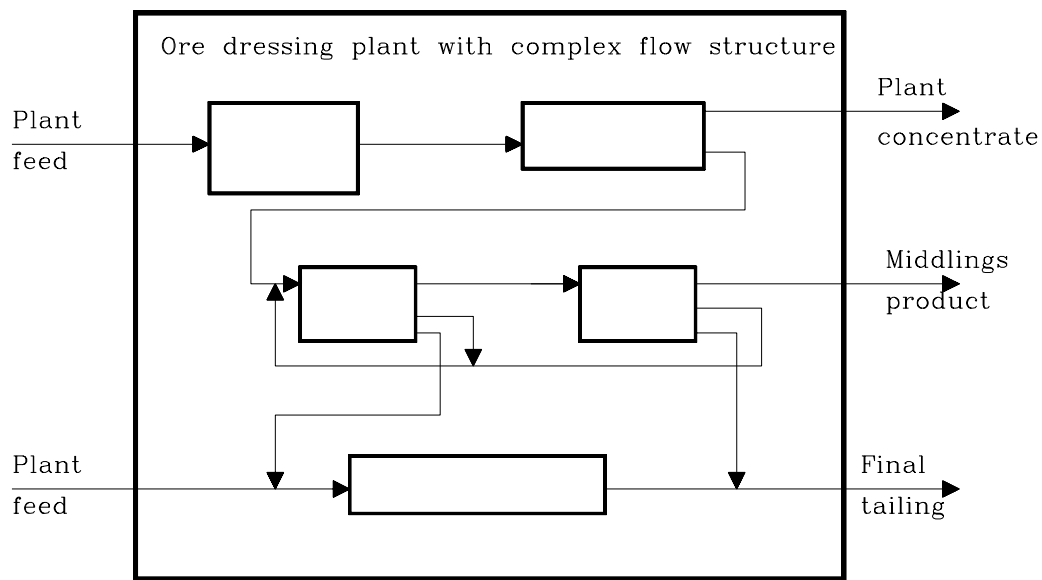


Figure 2.3 *Schematic representation of the complex structure of an ore-dressing plant.*

product. However, because of the recycle streams in the plant, it is not possible to calculate the composition of the product streams by moving sequentially from one unit to the next.

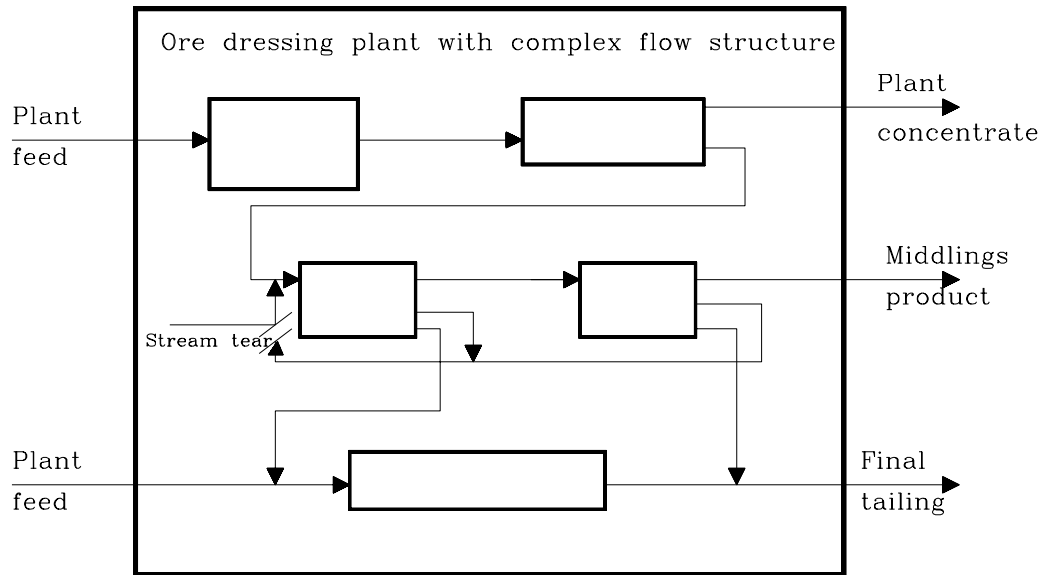


Figure 2.4 *Illustration of stream tearing to render the plant acyclic.*

However the flowsheet can be rendered acyclic by appropriate tearing. To achieve this some streams are torn open to break open all the loops in the plant. This is illustrated in Figure 2.4.

The structure of the simulator calculation procedure can be seen most clearly using a concrete example. Consider a flotation plant that includes a regrind mill as shown in Figure 2.4. Streams 3 and 5 are selected as tear streams. Streams 11 and 12 be torn instead of stream 5 but this would lead to a total of three tear streams rather than two. In general it is better to use the smallest number of tear streams that will make the plant acyclic.

Let W_i represent the vector of mass flowrates of particle classes in stream i and let W_i' be the assumed value of the vector for tear stream i . The formal matrix-vector models of the units as given in equations(4.2.1) and (4.2.2) can be applied in series as follows

$$\begin{aligned}
\mathbf{W}_2 &= \mathbf{W}_{13} + \mathbf{W}_1 \\
\mathbf{W}_{13} &= \mathbf{E}_{T2} \mathbf{W}_3' \\
\mathbf{W}_2 &= \mathbf{E}_{T2} \mathbf{W}_3' + \mathbf{W}_1 \\
\mathbf{W}_4 &= \mathbf{E}_{C2} \mathbf{W}_3' \\
\mathbf{W}_3 &= \mathbf{T}_1 \mathbf{W}_2 = \mathbf{T}_1 \mathbf{E}_{T2} \mathbf{W}_3' + \mathbf{T}_1 \mathbf{W}_1 \\
\mathbf{W}_6 &= \mathbf{E}_{T3} \mathbf{W}_5' \\
\mathbf{W}_7 &= \mathbf{E}_{T4} \mathbf{W}_6 = \mathbf{E}_{T4} \mathbf{E}_{T3} \mathbf{W}_5' \\
\mathbf{W}_{12} &= \mathbf{E}_{C4} \mathbf{W}_6 = \mathbf{E}_{C4} \mathbf{E}_{T3} \mathbf{W}_5' \\
\mathbf{W}_8 &= \mathbf{E}_{C3} \mathbf{W}_5' \\
\mathbf{W}_9 &= \mathbf{E}_{T5} \mathbf{W}_8 = \mathbf{E}_{T5} \mathbf{E}_{C3} \mathbf{W}_5' \\
\mathbf{W}_{11} &= \mathbf{T}_6 \mathbf{W}_9 = \mathbf{T}_6 \mathbf{E}_{T5} \mathbf{E}_{C3} \mathbf{W}_5' \\
\mathbf{W}_5 &= \mathbf{W}_4 + \mathbf{W}_{12} + \mathbf{W}_{11} \\
&= \mathbf{E}_{C2} \mathbf{W}_3' + \mathbf{E}_{C4} \mathbf{E}_{T3} \mathbf{W}_5' + \mathbf{T}_6 \mathbf{E}_{T5} \mathbf{E}_{C3} \mathbf{W}_5' \\
\mathbf{W}_{10} &= \mathbf{E}_{C5} \mathbf{W}_8 = \mathbf{E}_{C5} \mathbf{E}_{C3} \mathbf{W}_5'
\end{aligned} \tag{2.4}$$

The feed to the plant consists of a single stream (stream 1) and is represented by the vector

$$\mathbf{F}_1 = \mathbf{W}_1 \tag{2.5}$$

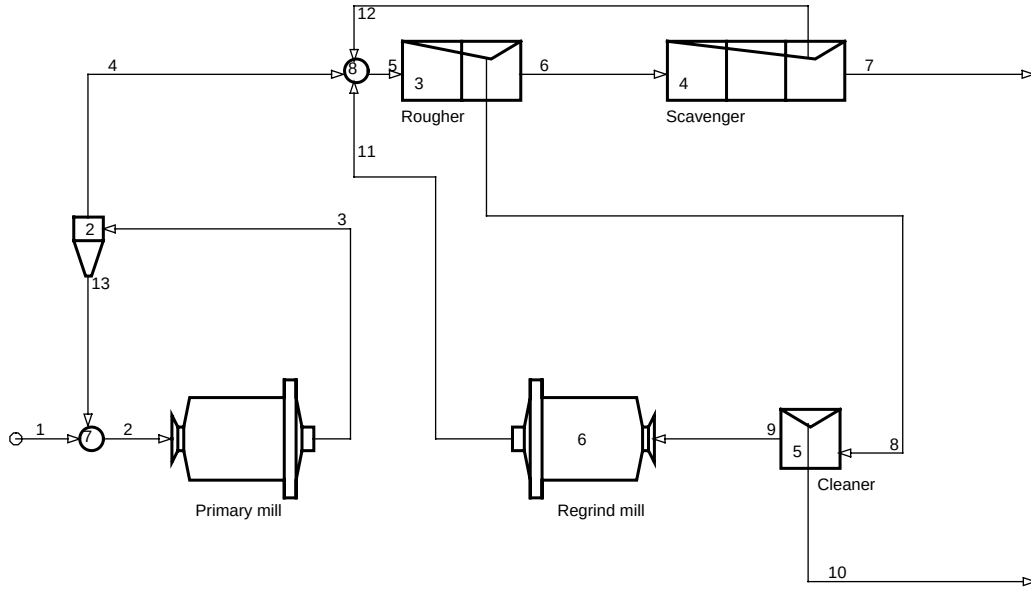


Figure 2.5 An ore-dressing plant flowsheet showing a flotation plant that includes a regrind mill.

The plant has two product streams (streams 7 and 10) and the product is represented by the composite vector

$$\mathbf{P} = \begin{bmatrix} \mathbf{W}_7 \\ \mathbf{W}_{10} \end{bmatrix} \quad (2.6)$$

The two tear streams are combined to form a composite vector

$$\mathbf{Y} = \begin{bmatrix} \mathbf{W}_3 \\ \mathbf{W}_5 \end{bmatrix} \quad (2.7)$$

and the calculated values of the tear streams

$$\mathbf{X} = \begin{bmatrix} \mathbf{W}_3' \\ \mathbf{W}_5' \end{bmatrix} \quad (2.8)$$

The operation of the plant is concisely described by the two matrix equations

$$\mathbf{Y} = \Phi \mathbf{X} + \Psi \mathbf{F} \quad (2.9)$$

and

$$\mathbf{P} = \Xi \mathbf{X} + \Theta \mathbf{F} \quad (2.10)$$

These matrices are composites formed from sums of products of the efficiency and transformation matrices of the units in the plant as shown below

$$\Phi = \begin{bmatrix} \mathbf{T}_1 \mathbf{E}_{T2} & \mathbf{0} \\ \mathbf{E}_{C2} & \mathbf{E}_{C4} \mathbf{E}_{T3} + \mathbf{T}_6 \mathbf{E}_{T5} \mathbf{E}_{C3} \end{bmatrix} \quad (2.11)$$

Each component matrix on the diagonal of matrix Φ represents the paths from the tear stream back to itself. It is easy to see from Figure 2.5 that the path from tear stream back to itself passes through the tailings of unit 1 and the tailings of unit 2. Unit 1 is a transformation unit while unit 2 is a separation unit and therefore the diagonal composite matrix is the product $\mathbf{T}_1 \mathbf{E}_{T2}$. On the other hand, two paths connect the tear in stream 5 back to itself: one passes through the tailings of unit 3 and the concentrate of unit 4 and the other through the tailings of unit 6, the tailings of unit 5 and the concentrate of unit 3. The mixing point at unit 8 is accounted for by adding the two product terms that represent each of the distinct paths. This path structure is reflected in the lower right diagonal of the matrix Φ . Likewise the off-diagonal component matrices of Φ represent paths in the flowsheet from one tear to another. Note that, in Figure 2.5, there is no direct path from tear 5 back to tear 3 hence the zero matrix on the upper right position. There is a direct path from tear 3 to tear 5 through the concentrate of unit 2 hence the matrix $\mathbf{E}_{C2}$ in the lower left position of Φ .

$$\Psi = \begin{bmatrix} T_1 \\ \mathbf{0} \end{bmatrix} \quad (2.12)$$

$$\Xi = \begin{bmatrix} \mathbf{0} & E_{T4} E_{T3} \\ \mathbf{0} & E_{C5} E_{C3} \end{bmatrix} \quad (2.13)$$

$$\Theta = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix} \quad (2.14)$$

The matrix Ξ is constructed in the same way as matrix Φ except that the elements of Ξ track direct paths from the tear streams to the products. In this example there are two such paths from tear 5; one through the tailings of units 3 and 4 and one through the concentrates of units 3 and 5. There are no direct paths from tear 3 to any product stream - all paths from tear 3 to the products pass through another tear. Hence both component matrices in the left-hand column of Ξ are zero.

The component matrices of Ψ represent direct paths from the plant feeds to the tear streams while the component matrices in Ξ represent direct paths from the plant feed streams to the product. In the present example, no paths connect the plant feeds to the plant products without passing through a torn stream. Hence both elements of Θ are zero in this example.

Algorithms for the identification of all loops in a complex plant are well known using standard theorems of mathematical graph theory and all loops can be reliably identified. The choice of which streams should be torn is not uniquely defined although the set of tear streams should be chosen to ensure that the iterative calculation procedure converges as efficiently as possible. MODSIM has excellent loop-finding and stream tearing algorithms built in so that the user of these systems never has to be concerned about this aspect of the problem.

The simulation problem is solved as soon as a vector is found that satisfies equation (4.2.9).

$$\mathbf{X} = \mathbf{Y} = \Phi \mathbf{X} + \Psi \mathbf{F} \quad (2.15)$$

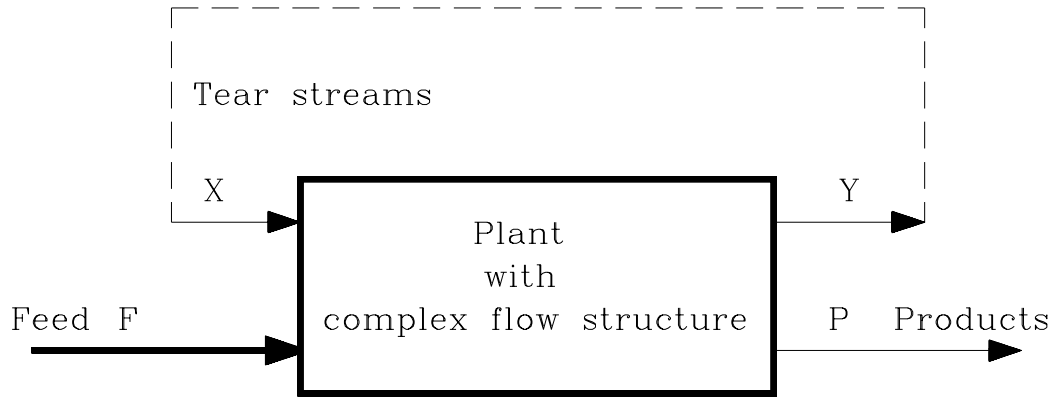


Figure 2.6 Schematic representation of equation (12).

The nature of this description of the plant is shown in Figure 2.6.

The development of solution procedures is a specialized task and two methods of solution have been considered: matrix inversion and iteration.

A formal solution by matrix inversion is easy to develop.

$$\mathbf{X} = \Phi \mathbf{X} + \Psi \mathbf{F} \quad (2.16)$$

$$\mathbf{X}[\mathbf{I} - \Phi] = \Psi \mathbf{F} \quad (2.17)$$

$$\mathbf{X} = [\mathbf{I} - \Phi]^{-1} \Psi \mathbf{F} \quad (2.18)$$

From which the vector of product stream can be evaluated.

$$\mathbf{P} = \Xi \mathbf{X} + \Theta \mathbf{F} \quad (2.19)$$

The method is attractive because only the terminal and tear streams occur in the description and all solution calculations can be completed on these streams. Other streams can be regenerated after solution is achieved. Thus a great economy of description has been achieved. In the case of a linear plant Φ is a constant matrix.

The apparent simplicity of the method is deceptive however. Often Φ is not constant and many of its elements depend on the solution vector X in a non-linear and complex fashion. Furthermore algorithms to calculate all of the elements of the component matrices are difficult to develop for general plant configurations. Direct iterative methods have become popular as a result. In this method, the numerical elements of the unit transformation and efficiency matrices are not evaluated explicitly. Rather the model of each unit operation is coded to accept the unit feed and produce a description of the unit product or products. In other words the transformation and efficiency matrices for the units exist only as computer code (often extremely complex) that defines the transformation from feed to product for each unit in the plant. The various paths from plant feeds and torn streams to the tears are tracked sequentially - the product from one unit being passed along as feed to the next unit in the path. The calculation must be started by making assumptions regarding the nature of the torn streams. The composition finally calculated at each tear is compared with the original assumption and the initial assumptions are improved by iteration.

Various methods for the iterative solution are available.

I. Direct substitution

$$X^{(k+1)} = \Phi X^{(k)} + \Psi F \quad (2.20)$$

where $X^{(k)}$ is the k th iterate towards the solution and Φ and Ψ may also be functions of $X^{(k)}$.

ii. Newton-Raphson

$$X^{(k+1)} = X^{(k)} - (I - \Phi'(X))^{-1} (X^{(k)} - \Phi X^{(k)}) \quad (2.21)$$

where $\Phi'(X)$ is the matrix of derivatives of $\Phi(X^{(k)})$ with respect to the elements of X . In practice $\Phi'(X)$ is difficult to obtain and various approximations must be used. MODSIM includes efficient algorithms to ensure rapid and reliable convergence of this iterative calculation. MODSIM, in particular, includes an efficient algorithm for the approximation of Φ' and it gives rapid convergence in most practical problems.

There is no guarantee that a solution to equation (4.2.15) does in fact exist and, if a solution does exist, that it is unique. Many model equations that implicitly generate the elements of the transformation and efficiency matrices can yield an overall matrix Φ that does not allow a solution to equation (4.2.15). Unfortunately, it is not possible to predict *a priori* when a solution does not exist and this only shows up in practice when the iterative scheme fails to converge.

CHAPTER 3

SIMULATION OF ORE DRESSING PLANTS - SIMPLIFIED MODELING TECHNIQUES

3.1 Comminution units

The fundamental principles that underlie modern developments in the modeling of comminution and classification units will be presented in this section. No attempt will be made to comprehensively cover the details that must be incorporated in the model and only the basic concepts will be explained. This will be sufficient for the reader to understand how these models are used in simulation studies and how they can be used to interpret experimental data obtained from operating laboratory, pilot plant or full-scale equipment.

Both crushing and grinding operations are transformation processes as defined in the previous chapter. These units transform the characteristics of the feed material in the sense that the particle size distribution of the feed material is drastically changed during passage through the unit. A grinding mill will also give rise to a significant change in the liberation spectrum the mineralogical material because grinding will change the size of the solid particles on a scale that is comparable with the mineralogical grain size of the ore that is processed. Indeed grinding operations are generally included in an ore dressing plant specifically to liberate the valuable minerals from the ore. In terms of the fundamental description of the particulate state, the comminution units transfer material from one particle class in the feed to a different particle class in the product and simultaneously material transfers from one grade class to another. This is in sharp contrast to separation units in which any solid particle remains in the class that it occupied in the feed but is directed to one or other of the output streams from the unit. For example, a vibrating screen does not change the size of the solid material that it processes but directs it to overflow or underflow depending on whether the size is greater or less than the mesh opening in the screen.

Comminution units must necessarily be single-input single-output units because they do not attempt to achieve any separation of the particles. Any mathematical model of a comminution unit must define the manner in which the unit transforms the particulate nature of the feed into the particulate structure of the product material.

3.2 Crushing Units

The basic operation of a crushing machine is rather simple. Large pieces of solid material enter the crushing region of the machine during the period of the crushing cycle when the machine is relatively open. The machine then enters a period when the crushing zone is relatively closed. The forces imposed by the machine are sufficient to fracture the solid. During the next open period of the cycle two things happen. New material enters the crushing zone and pieces crushed on the previous cycle nip move deeper into the crushing zone or fall completely through it and are discharged in the product stream. Obviously the smaller the particle the greater the chance that it will fall through the crushing zone and enter the product stream.

If a particle is broken during the crushing stage it will produce fragments that are smaller than the parent particle. The distribution of sizes among the daughter fragments is primarily a function of the material that is being crushed. The distribution of sizes among the daughter products is of fundamental importance in any comminution operation. It is described quantitatively by the breakage function $B(x,y)$ which is defined as the fraction of material in the products that is smaller than x from the breakage of a single particle of size y .

The probability that a particle will fall completely through the crushing zone of the machine during the opening part of the cycle is described by the classification function $C(x)$. This is defined as the fraction of particles of size x which are retained in the crushing zone of the machine during the open part of the cycle.

Consider a single pair of crushing and opening stages in the crusher cycle and let us establish the material balance for a typical particle size class. Let the representative size for the class be given by d_i and the upper and lower boundaries of the size class be given by D_{i-1} and D_i .

It is usual to use a geometric classification on the size scale but this is not essential.

During the opening phase in the cycle, an amount F_i ($i = 1, 2 \dots NBC$) of material of representative size d_i enters the crushing zone from the crusher feed. Also material in size class i enters the crushing zone from breakage of all other sizes larger than size D_{i-1} during the breaking stage of the previous cycle. Let M_j represent the amount of material in size class j retained in the crushing zone. A mass balance on material in size class i gives

$$M_i = (F_i + \sum_{j=1}^i M_j \Delta B_{ij}) C_i \quad (3.1)$$

In equation (3.1) the symbol ΔB_{ij} is given by

$$\Delta B_{ij} = B(D_{i-1}, d_j) - B(D_i, d_j) \quad (3.2)$$

and is equal to the fraction of material that appears in size class i from fracture of material in size class j . The multiplying factor C_i in equation (3.2) reflects the fact that only a fraction of the material entering the crushing zone in size class C_i remains to be broken, the rest passing through to be discharged in the crusher product.

The amount of material of size i entering the crusher product during the opening phase is given by

$$P_i = (F_i + \sum_{j=1}^i M_j \Delta B_{ij})(1 - C_i) = M_i \frac{(1 - C_i)}{C_i} \quad (3.3)$$

Equation (3.3) is the fundamental modeling equation for crushers and can be effectively used for digital simulation because it permits an easy solution by successive substitution.

A computer algorithm is easy to develop as follows:

$$M_i = F_i C_i + C_i \sum_{j=1}^{i-1} M_j \Delta B_{ij} + C_i M_i \Delta B_{ii} \quad (3.4)$$

$$\Delta B_{ii} = B(D_{i-1}, d_i) - B(D_i, d_i) = 1 - B(D_i, d_i) \quad (3.5)$$

ΔB_{ii} represents the fraction of the daughter material that remains in size class i after the crushing.

From equation (3.4)

$$M_i(1 - C_i \Delta B_{ii}) = F_i C_i + C_i \sum_{j=1}^{i-1} M_j \Delta B_{ij} \quad (3.6)$$

$$P_i = \frac{M_i (1 - C_i)}{C_i} = \frac{F_i(1 - C_i) + (1 - C_i) \sum_{j=1}^{i-1} M_j \Delta B_{ij}}{1 - C_i \Delta B_{ii}} \quad (3.7)$$

Equations (3.6) and (3.7) are easy to solve recursively starting with $i = 1$ the class having the largest representative size.

This general model framework is very powerful and only requires the classification function and the breakage function to be determined for each type of crushing machine. It is believed that the breakage function can be established adequately from single particle crushing tests while the classification function is obviously related to the geometry of the crushing machine. An assumption that is often made is that the breakage function is normalized so that it is a function only of the ratio D_i/D_j . Typical forms for the breakage and classification functions are shown in Figure 3.1.

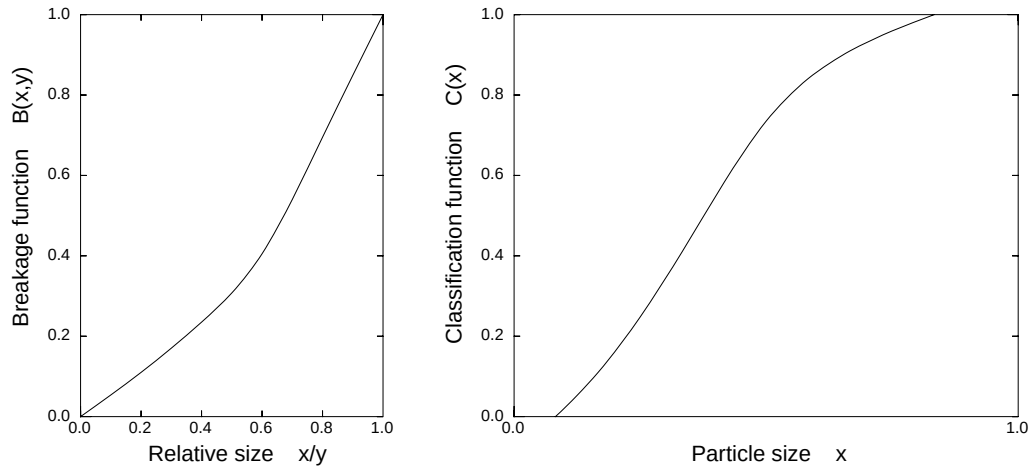


Figure 3.1 Typical forms for the breakage and classification functions for crushing machines.

The classification function must obviously have the characteristic that $C(x) = 0$ whenever x is much less than the closed side setting and $C(x) = 1$ whenever x is much greater than the open side setting. Typically $d_1 \approx 0.7 \cdot \text{CSS}$ and $d_2 \approx \text{OSS}$. The breakage function is largely dependant on the material characteristics while the classification function is dependent strongly on the design of the crushing machine.

The investigation of the nature of these functions represents the major area of research interest in comminution modeling at the present time. Unfortunately this course is too short to permit any detailed discussion of these very interesting questions.

3.3 Grinding Operations

The modeling of grinding machines such as ball, rod and pebble mills follows very similar lines to the crusher models that have just been discussed. However there are some significant differences and these are discussed here.

The material inside a grinding mill is broken continuously because of the action of the grinding media. A fundamental postulate made in establishing a model for grinding machines is that the rate of breakage is proportional to the amount of material of a specific type that is present in the grinding machine. In particular it is postulated that the rate of breakage of material of a particular size is proportional to the amount of material of that size present in the mill.

If M_i is the mass of material in size class i in the mill, then the rate at which that material breaks is given by

$$\text{Rate of breakage} = s_i M_i.$$

The constant of proportionality, s , is a function of the material in the mill (brittle materials have high values of s , tough materials low values of s of the representative size for class i , and of the environment in the mill such as the ball load, ball size, pulp viscosity and so on. The constant s is usually referred to as the selection function.

The result of breakage in the mill can be modelled by means of a breakage function just as in the crusher.

A material balance on size class i is easy to establish using the variables

F_i = mass flowrate of material in size class i in the feed

P_i = mass flowrate of material in size class i in the mill product

$$P_i = F_i - s_i M_i + \sum_{j=1}^i s_j M_j \Delta B_{ij} \quad (3.8)$$

An important simplification results if the working variables are changed to the fractions in a particular size class. Thus let

f_i = fraction of feed stream in size class i

p_i = fraction of product stream in size class i

m_i = fraction of mill charge in size class i

Equation (3.8) now becomes

$$Pp_i = Ff_i - s_i Mm_i + \sum_{j=1}^i s_j Mm_j b_{ij} \quad (3.9)$$

F and P represent the total flow of material in the feed and product streams and M is the mass of solid charge in the mill. Obviously $P = F$ and an important operating parameter is the average residence time that the solid material spends in the mills.

$$\text{Mean residence time} = \tau = \frac{M}{F} = \frac{M}{P} \quad (3.10)$$

$$p_i = f_i - s_i \tau m_i + \sum_{j=1}^i s_j \tau m_j b_{ij} \quad (3.11)$$

Equation (3.11) is the fundamental model equation for the continuously operating mill. This equation cannot be solved until some relationship is established between p_i and m_i . Such a relationship is determined by the flow pattern of solid material through the mill. This is a very complex problem and has attracted considerable research interest. However we cannot pursue this very interesting problem here and we use an assumption that is very commonly employed for practical modeling studies. If the material inside the mill is perfectly well mixed then the particle size distribution in the product is exactly the same size as that throughout the mill and then

$$p_i = m_i \quad (3.12)$$

This assumption immediately leads to a solution of equation (3.11) as follows

$$p_i = \frac{f_i + \sum_{j=1}^{i-1} s_i \tau b_{ij} p_j}{1 + s_i \tau} \quad (3.13)$$

Equation (3.13) is particularly suitable for digital computer evaluation because it leads at once to a very convenient evaluation by recursion starting from $i=1$. Thus the particle size distribution in the product is evaluated from the particle size distribution in the feed.

This basic structure is suitable for a wide range of modeling applications particularly simulation. However to be useful it is necessary to have effective model equations for the breakage function B and for the selection function s . Considerable information exists in the literature regarding such models but space does not permit detailed discussion here.

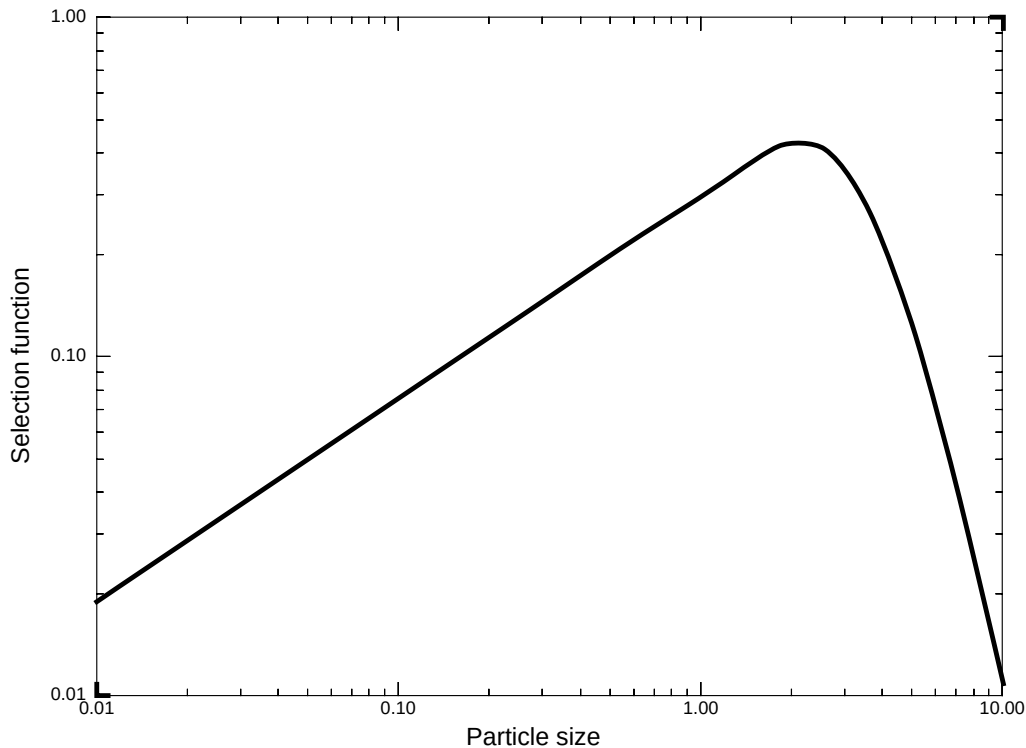


Figure 3.2 A typical selection function for ball milling plotted using equation (3.14).

The selection function is a strong function of particle size and a typical functional form is shown in Figure 3.2. The selection function is also strongly influenced by the size of the mill and the milling

environment inside the mill. Thus the selection function can be effectively used to define the scale-up characteristics for the mill.

A commonly used functional form for the selection function is that developed by Austin

$$s_i = a(d_i)^\alpha Q_i \quad (3.14)$$

$$Q_i = \frac{1}{1 + (d_i/\mu)^\Lambda} \quad \Lambda \geq 0 \quad (3.15)$$

The factor Q_i models the decrease in the rate of breakage as the particle size increases to approach that of the media. A functional form for the breakage function that is useful for both crushing and grinding models is given by

$$B(x,y) = \Phi(x/y)^\gamma + (1-\Phi)(x/y)^\beta \quad \text{with } 0 \leq \Phi \leq 1 \quad (3.16)$$

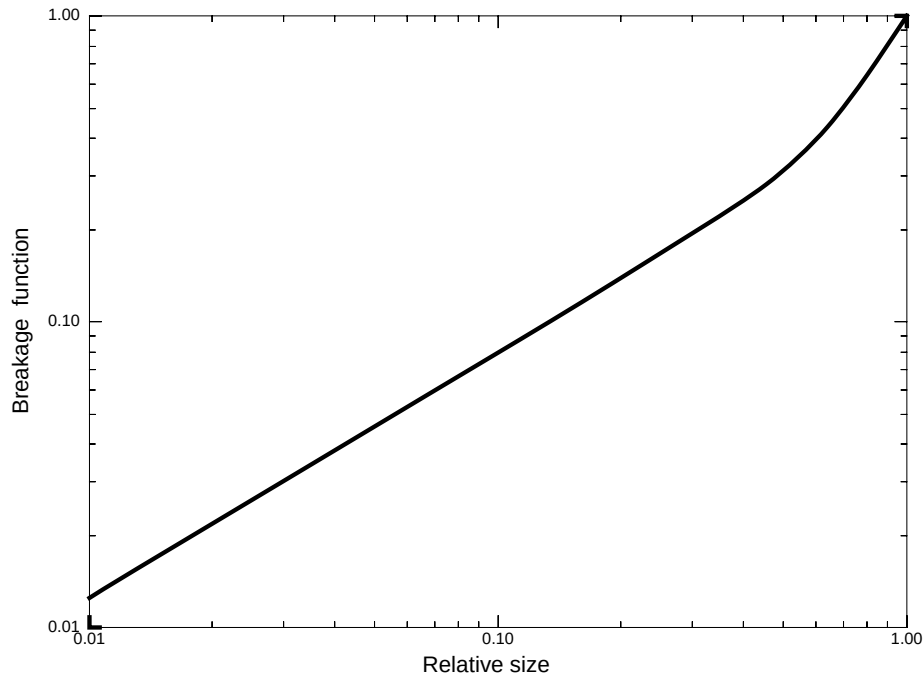


Figure 3.3 A typical breakage function plotted according to equation (16).

The variable Φ represents the fraction of fines in the progeny particles that result from a single fracture event.

The selection and breakage functions must be determined for each particular modeling application and it is through these functions that the general comminution models that have been described above can be made to describe the behavior of individual comminution operations. Often it is completely adequate to adjust the parameters in the simple functions shown in equations (3.14) and (3.16) using data appropriate to the equipment that is to be modelled. Usually such data is obtained in a laboratory batch mill and then the parameters in equations (3.14) and (3.16) must be scaled up to suit the full-scale operation. Good scale up procedures have been developed and it is now possible to predict full-scale performance from laboratory or pilot plant data. For very accurate modeling more elaborate functional forms for selection and breakage functions must be used. Unfortunately time and space do not allow us to investigate these questions here.

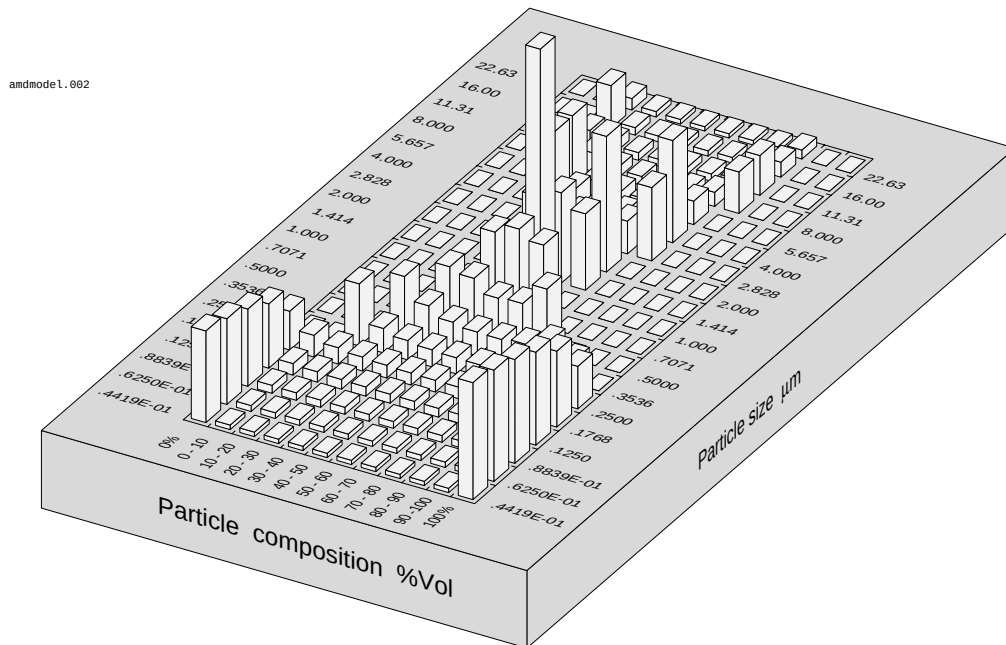


Figure 3.4 A typical Andrews-Mika diagram which shows the disposition of the progeny particles when a parent of known composition and size is broken.

The question of liberation of minerals during grinding must be addressed here. When liberation is an important aspect of the processing of a particular ore, the models used for the comminution operations must be modified to incorporate this effect. The method used is based on the classic analysis of this problem by Andrews and Mika in 1975. The essence of this method is the modeling of the distribution of progeny particles among the grade classes as well as the particle size classes

when a parent particle of known composition is broken. The breakage function b_{ij} must be replaced by a different function which has as its arguments the composition and size of the parent particle and the composition and size of the progeny particles. For two-component mineral textures, this function can be represented graphically as shown in Figure 3.4.

This diagram shows what distribution of progeny a parent particle consisting of 50% - 60% mineral phase would produce during comminution. It is necessary to build up a diagram such as this for every possible parent particle composition and size. MODSIM includes two excellent models for constructing these diagrams.

3.4 Classification, dense-medium, magnetic separation and flotation units

The second important group of ore dressing unit operations are the separation units. These operate on the basis of separating particles according to their size or composition. The principles of most separation units are similar in that the particles having the desired characteristics are directed towards the appropriate product stream. All separation units have one characteristic in common - the separations are not perfect and to a larger or lesser extent particles of the wrong type find their way into the outlet streams. This is termed misplaced material and gives rise to the loss of performance of the ore dressing plant as a whole with the loss of valuable material in the plant tailings and the recovery of undesirable material in the concentrate.

Generally the behavior of a particular solid particle in a separation unit will depend on both its composition and its size often leading to separations with respect to one property being confounded by the separation due to another. All separation units have some dominating property on which the separation is based and this can be particle size, particle density or particle composition. As examples of these unit separations we will consider classifiers, dense-medium separators and flotation cells.

3.5 A Generalized Model for Separation Units.

It is possible to establish a generalized model for separation units that provides a framework for many of the specific units that are in common use. Such a generalized model reflects the important operating characteristics of the units that ultimately define their operating performance. Most separation units have four major performance indicators and any model should be able to account for each of those if it is to provide a satisfactory description of the unit operation. The four performance indicators are

- the cut point

- the separation efficiency
- the short circuit flow to tailings
- the short circuit flow to concentrate

The cut point

In the case of classifiers and gravity separation units, a specific cut point is defined which theoretically defines the operating size or specific gravity which separates the material into the two product streams. Ideally all material should split according to the cut point and the cut point is the chief operating variable that determines the nature of the two product streams that are produced in the separation.

The separation efficiency

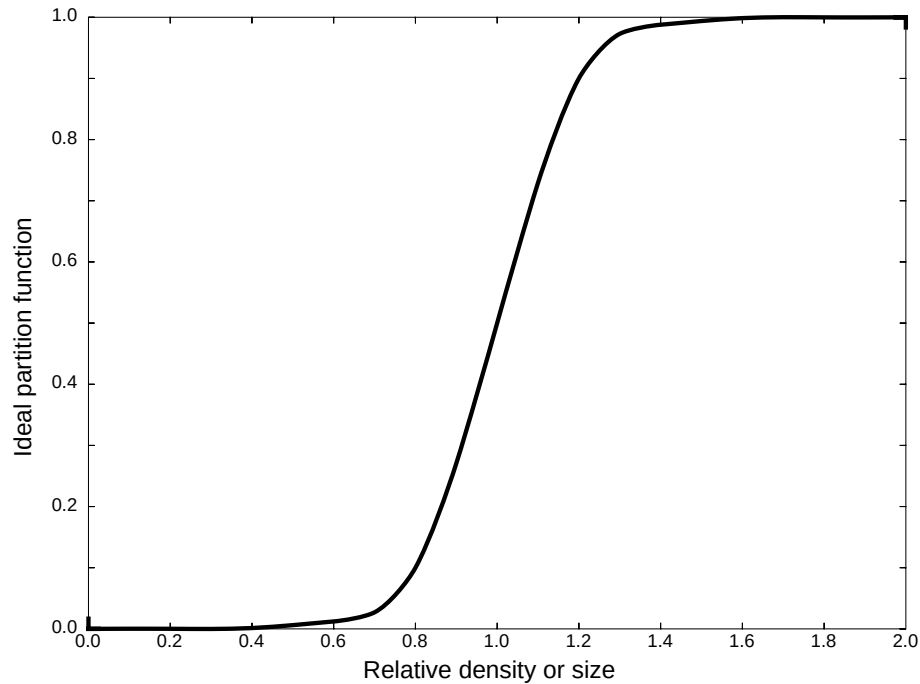


Figure 3.5 *Ideal or corrected partition curve for a separation unit.*

Ideally particles should be separated at the cut point but this is never perfectly achieved in practice because a certain number of the particles will follow the incorrect path through the unit and end up in the wrong outlet stream. The more efficient a separation unit is, the less the chance of a particular particle being misplaced into the wrong outlet and all separation units are designed to achieve as efficient a separation as possible. Various measures of separation efficiency have been proposed and used in practice and it is necessary to model this efficiency to describe the operation of real separation units.

The cut point and the separation efficiency are commonly described by reference to a partition curve as drawn in Figure 3.5.

The partition factor shows the fraction of material in the feed that reports to the heavy product. In the case of a classifier this would be the oversize and in the case of a gravity separation unit this

would be the heavy product. It is usual and very useful to plot the partition curve against the normalized separation variable, that is the ratio of the separation variable to the cutpoint. It is also usual to specify the cutpoint as the value of the separation variable at which the corrected partition function is exactly 0.5. This normalization in many cases produces a generalized curve that is specific to the unit and is independent of the actual cutpoint that is set for the separation. The separation efficiency is a measure of the sharpness of separation at the cutpoint.

The short circuit flows

In many separation unit operations a certain proportion of the feed material passes directly to one or other of the product streams without undergoing any form of separation. This has a very significant effect on the performance of the unit and must be allowed for if the behavior of the unit is to be correctly modelled. The effect of these short circuit flows can be evaluated by considering the schematic diagram of a separation unit as shown in Figure 3.6.

Let α and β represent the fraction of the feed that short circuits directly to tailing and concentrate respectively. A material balance on solid material of particular type gives

$$C_i = \beta F_i + e_i F_i (1 - \alpha - \beta) \quad (3.17)$$

where e_i is the ideal or corrected partition factor for particles of type i . C_i is the flowrate of particles of type i in the concentrate and F_i the flowrate of particles of type i in the feed.

$$E_i = \frac{C_i}{F_i} = \beta + (1 - \alpha - \beta)e_i \quad (3.18)$$

This actual partition factor will be a function of the separation variable as shown in Figure 6.

Several measures of separating efficiency are commonly used and most of these can be related to the

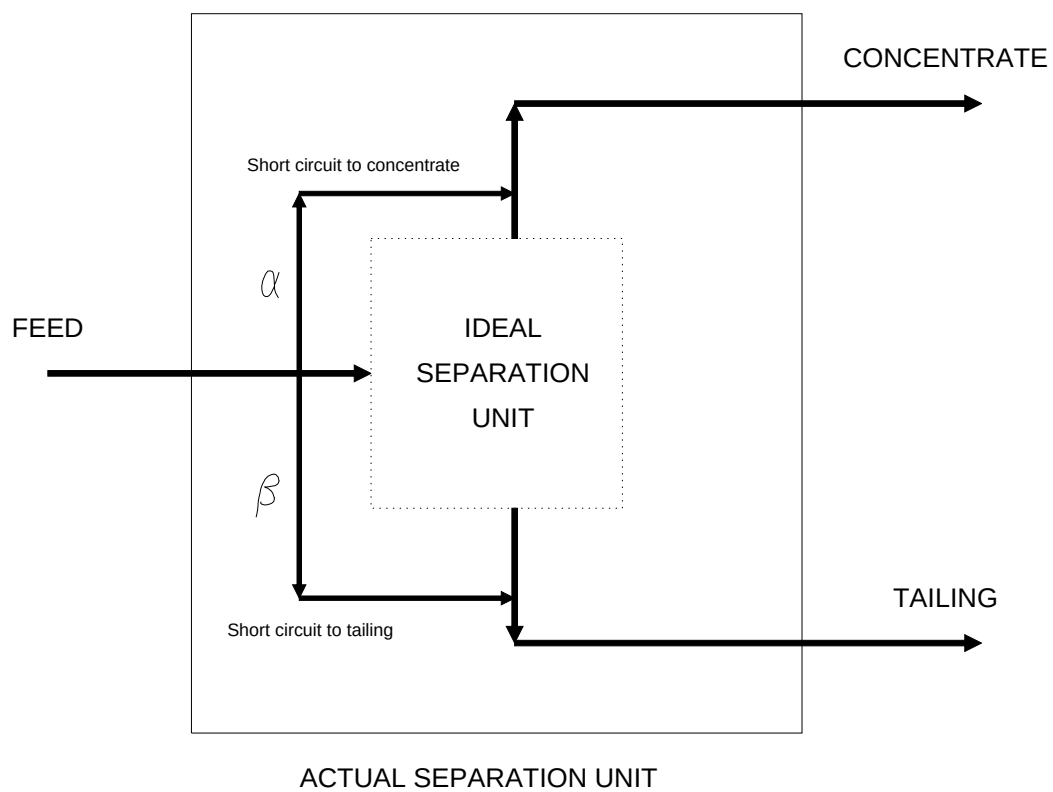


Figure 3.6 *Stylized representation of a real separation unit that is subject to short-circuit flows.*

shape of the partition curve in the region close to the cutpoint. Two measures are particularly common, namely the imperfection and the separation index. Both are related to the point at which the corrected or ideal partition curve passes through the 0.75 and 0.25 points. Let x_{25} be the normalised value of the separation variable at the point where the partition function has the value 0.25 and a similar definition for x_{75} . Then the imperfection is defined by

$$I = \frac{x_{75} - x_{25}}{2} \quad (3.19)$$

and the sharpness index is defined by

$$SI = \frac{x_{25}}{x_{75}} \quad (3.20)$$

Empirical Models for Corrected Partition Function.

It is a very difficult task to develop models for the partition curve because of the enormously complex behavior patterns exhibited by solid particles in the various devices that are used in ore dressing plants. Most successful models are based on one or other empirical functional forms for the corrected partition function with each of the four performance indicators related to the type and size of unit and to the nature of the particles. For example, in a hydrocyclone the cutpoint will vary significantly with cyclone diameter and with particle specific gravity but the shape of the corrected partition curve will not vary in cyclones that are geometrically similar. Similarly in a dense medium cyclone the cutpoint specific gravity will be a strong function of the medium density and the particle size but only a weak function of the cyclone size. However, the imperfection is a function of particle size and cyclone diameter.

A considerable amount of information is available in the literature on the variation of these four performance indicators for a wide variety of separation units.

Some empirical functions that have been found useful in practice to describe the corrected partition curve are

$$e_i = \frac{e^{\lambda x_i} - 1}{e^{\lambda x_i} + e^{\lambda - 2}} \quad (3.21)$$

where

$$\lambda = \frac{1,099}{I} \quad (3.22)$$

or

$$SI = \frac{\ln[(e^{\lambda} + 2)/3]}{\ln[3e^{\lambda} - 2]} \quad (3.23)$$

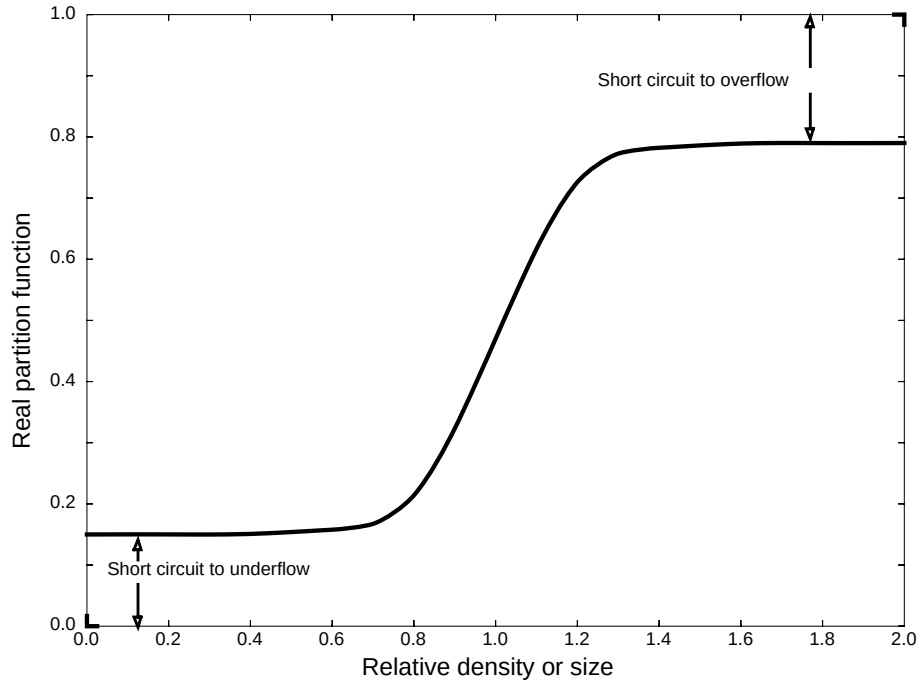


Figure 3.7 Actual partition curve for real separation units that are subject to short-circuit flows. Curve was drawn using equation () in equation (). The effect of the short-circuit flows is clearly shown.

Another empirical corrected partition function is

$$e_i = 1 - \exp(-0,693x^\lambda) \quad (3.24)$$

where

$$I = [e^{0,6951/\lambda} - e^{-0,8792/\lambda}] \quad (3.25)$$

or

$$SI = e^{-1,5725/\lambda} \quad (3.26)$$

3.6 Flotation Cells

Although flotation cells are essentially separation devices it is not usual to model their performance by the partition function. The reason is that the separation is effected by differences in the rate at which the various particle types can be recovered by transfer from the pulp to the froth. Flotation is a kinetic process and is best modelled in terms of the rate of flotation.

The flotation process involves several microprocesses that must take place before a mineral particle can be recovered in the froth phase. These processes are

- * suspension of solid particles in the pulp phase
- * collision between a particle and a bubble
- * adhesion of the particle onto the bubble
- * lifting of the particle by the bubble into the froth phase
- * transmission of the particle through the froth layer to be recovered in the concentrate stream.

Any model that is used to describe the flotation process should account for these processes and relate them to the nature of the particles and the environment within the flotation cell. Of course, the nature of the solid surface is crucial in determining the adhesion characteristics on the bubble and the behavior of the solid particle in the froth.

Unfortunately it has not proved to be possible to develop reliable quantitative models for these individual microprocesses, particularly those that occur in the froth phase. However, empirical descriptions of the dominating processes have proved to be moderately successful and an empirical rate model can be used.

Rate of transfer of particles in class i from pulp to froth = $k_i S M_i$

where M_i is the mass of particles of class i in the pulp in the cell, S is the available bubble surface area per unit of pulp volume and k_i is an inter phase transfer rate constant for transfer of solid from the pulp to the froth phase.

Not all of the solid particles that enter the froth phase are recovered in the concentrate stream. A certain fraction fall back into the pulp before they pass over into the concentrate. This effect has been modelled in a variety of ways and the result is the calculation of a fractional transmission efficiency for the froth, such that

Rate of recovery of particles in class $i = \gamma_i k_i S M_i$ where γ_i is the froth transmission efficiency.

This empirical kinetic model must be combined with a hydrodynamic model of the pulp phase to calculate the recovery of a particular type of particle. Most flotation cells are very well mixed because of the intense agitation and aeration required so that the assumption of perfect mixing is usually adequate.

Let f_i , t_i , c_i represent the mass fractions of particles in class i in the feed, tailing, concentrate streams and in the pulp phase in the cell respectively. The material balance for material in class i becomes

$$F f_i = \gamma_i k_i S M m_i + T t_i \quad (3.27)$$

In this equation F and T represent total flow rate of solid in the feed and tailings and M represents the mass of solid in the pulp phase in the cell.

$$\begin{aligned} \text{Recovery of particles in class } i &= 1 - \frac{T t_i}{F f_i} \\ &= \frac{\gamma_i k_i S M}{T + \gamma_i k_i S M} \end{aligned} \quad (3.28)$$

In deriving equation (3.28) the fact that the cell is assumed to be perfectly mixed allows us to use $m_i = p_i$. In other words the composition of the solid phase in the tailing is the same as the composition throughout the cell. The residence time of solid in the cell is given by

$$\tau = \frac{M}{T} \quad (3.29)$$

Thus

$$\text{Recovery of particles of type } i = \frac{\gamma_i k_i S \tau}{1 + \gamma_i k_i S \tau} \quad (3.30)$$

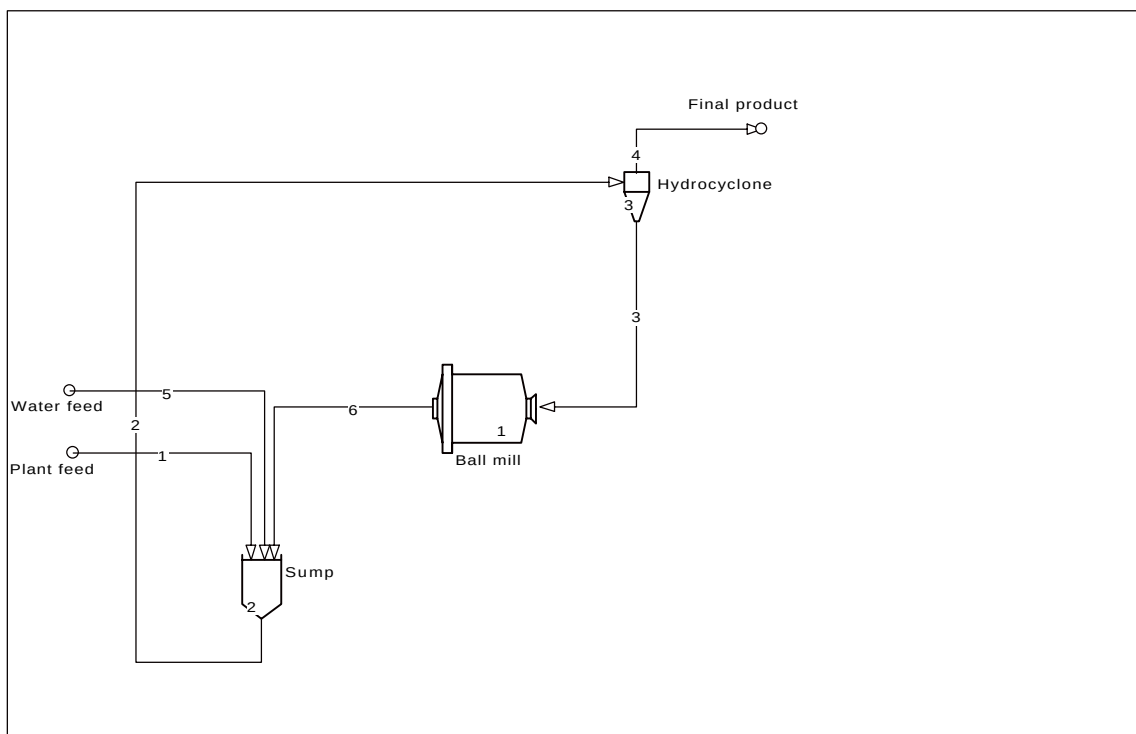
The modeling of the factors γ_i , k_i and S has received considerable attention in the literature and many subtle variations of equation (3.30) have been used to model flotation cells. In particular, particles of a particular size and composition exhibit a variety of rate constants k and this has led to the popular distributed constant flotation model. This model requires the distribution of particles over several k classes as well as over the size and grade classes.

MODSIM offers two variations of the kinetic model for flotation.

CHAPTER 4

EXAMPLES OF SOME SIMPLE SIMULATIONS

EXAMPLE 1 Ball mill circuit



A ball mill operates in closed circuit with a hydrocyclone classifier. It processes rod mill discharge at 800 tons solid per hour. The size distribution of the feed is the rod mill discharge given in Table 1 and the pulp is 76% solids. The circuit flowsheet is shown in the diagram. The hydrocyclone should be operated at 50% solids. The size of the mill will give a residence time of 7 mins. The ore can be taken as silica having a specific gravity of 2.67.

Draw the flowsheet using MODSIM and specify the necessary data.

Simulate the circuit and show the size distributions in the plant feed, the ball mill feed, the ball mill discharge and the circuit product. Investigate how the relationship among these four size distributions varies as the cutpoint in the cyclone and therefore the circulating load changes.

Investigate the effect of hydrocyclone diameter on the D_{50} cutsize.

Investigate the effect of the number of cyclones in the cluster on the D_{50} cutsize..

EXAMPLE 2 Flotation cell

The flotation constants associated with the ore described in Table 4 are $k_1 = 1.23 \times 10^{-2} \text{ s}^{-1}$, $k_2 = 5.12 \times 10^{-3} \text{ s}^{-1}$, $k_3 = 8.2 \times 10^{-4} \text{ s}^{-1}$, $k_4 = 0$

Simulate the operation of a single 4-cell rougher bank of cells. The feed pulp density is 34% solids and solid holdup in the cells is 450 kg solid/m³ of cell volume. Plot the grade-recovery relationship as cell volume varies from a m³ to 10 m³. The feed rate is 135 tons/hr. The effect of particle size on the flotation kinetics can be ignored and the unliberated particles may be assumed to contain 16.8% mineral on average. The feed contains 10% liberated mineral, 5% unliberated particles and 85% liberated gangue.

Note: The simple kinetic model with distributed rate constants should be used for the flotation cells. This is identified as model FLTN in MODSIM.

EXAMPLE 3 Aggregate Plant

A crushing and screening plant used for the production of building aggregate is shown in Figure 1. The equipment schedule is given in Table 1. The plant produces three products with the material from streams 12 and 15 being combined to produce a middling product.

The plant processes 400 tonnes/hr of feed and the size distribution in the feed is given in Table 2.

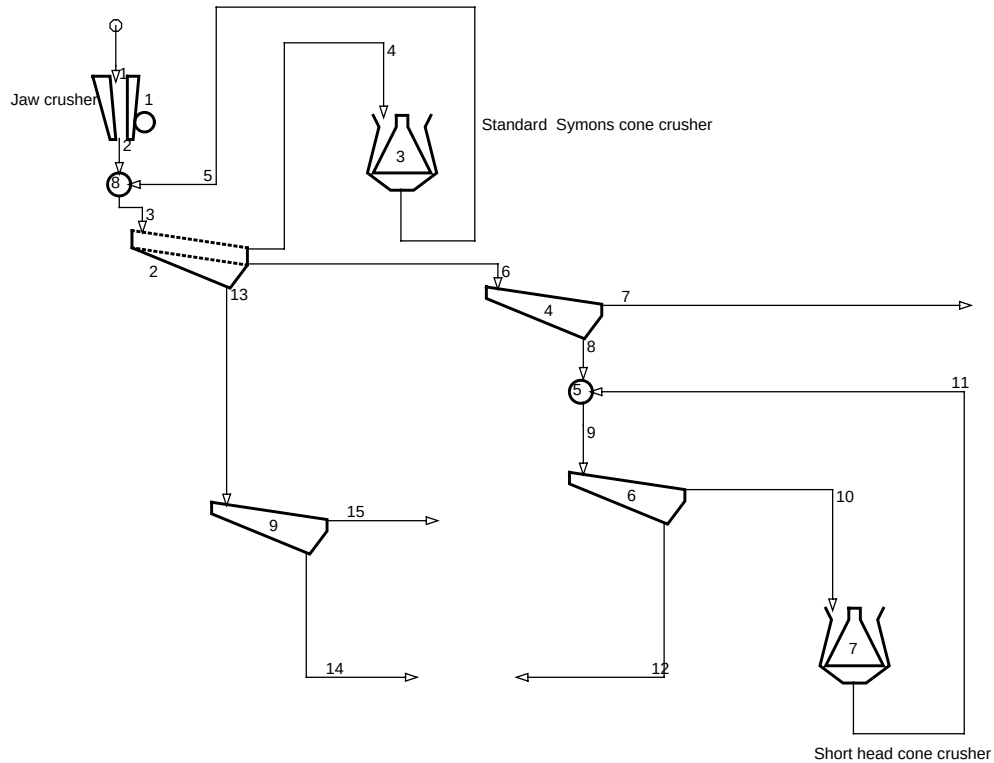


Figure 4.2 *Flowsheet of an aggregate plant*

Simulate the performance of this plant and investigate how the relative production rates of the three products, and their size distributions, vary as the closed side setting of the secondary crusher and the mesh size of the lower deck of the primary screen vary. You should be aware of any possible overload conditions that may occur as plant conditions are varied.

To display the results you should plot the production rates against the two control variables. To display the variations in the size distributions, plot the D_{80} , D_{50} and the D_{20} sizes against the mesh size of the screen.

Table 1 Equipment schedule for aggregate plant

Unit	Type	Settings
Primary crusher 1	Jaw crusher	2 × 630mm × 1000mm OSS 90mm
Primary screen 2	Double-deck inclined wire-mesh screen	1.83m × 3.66m 15° inclination Top deck: 90mm mesh opening 16mm wire Lower deck: 6mm mesh opening 3.4mm wire
Secondary crusher 3	Standard Symons cone	1.78m coarse cavity CSS 63mm
Secondary screen 4	Single-deck inclined wire-mesh screen	1.83m × 4.27m 15° inclination 25mm mesh 7.94mm wire
Tertiary screen 6	Single-deck inclined wire-mesh screen	1.52m × 4.27m 15° inclination 10mm mesh 4.1mm wire
Tertiary crusher 7	Short head cone	1.22m medium cavity CSS 6mm
Final product screen 9	Single-deck inclined wire-mesh screen	1.22m × 3.66m 15° inclination 1.5mm mesh

Table 2 Particle size distribution in the feed

Size mm	% passing
300	96
150	74
75	52
37.5	34
19.0	20
13.2	12
9.5	7
6.7	4
4.75	3
3.35	2
2.36	1

EXAMPLE 5 Effect of regrinding on the performance of base-metal sulphide flotation plants.

Introduction

The flotation behavior of composite materials is of considerable importance in the flotation of base-metal sulfides which usually have highly intergrown mineralogical textures. When these are ground, perfect liberation is never achieved so that locked particles are always present for subsequent separation processes. It is generally assumed that locked particles float in some manner intermediate between fast floating free particles and non-floating particles of gangue. It is believed that regrinding intermediate streams in a flotation plant will liberate some of the mineral and thus lead to better separation in subsequent flotation stages.

Little quantitative data is available to confirm this practice particularly because of the difficulty and cost of collecting data on the liberation spectrum in plant streams. It is only recently that techniques for quantitative mineralogical analysis have become available to make such investigations feasible.

The Mount Lyell Ore

Dr David Sutherland of the CSIRO in Australia has published an extensive study on the material from the Mount Lyell mine in Tasmania using the QEM*SEM technique for the examination of individual particles on a large scale. This excellent piece of work is published in references 1 and 2 and represents probably the most useful data set of its kind at present available.

The Sutherland study shows clearly that the flotation behavior of the Mount Lyell ore can be usefully modeled using the discrete distributed rate constant model with fast and slow floating components for each type of particle in the flotation feed. Their major conclusion is that the same flotation rate constants can be used for all particles containing the valuable mineral chalcopyrite but that as the gangue constant of the particle increases so the particles distribute more and more to the slower flotation rate constant.

At the same time techniques for the calculation of the liberation spectrum in the product from a continuously operating milling circuit have been developed by King³. It is possible now to combine the results of these two studies to simulate the effect of regrinding on the performance of a flotation plant.

In this study it is proposed to investigate a very simple plant consisting of a rougher and cleaner bank for the Mount Lyell ore. The plant with and without a regrind mill is shown in figures 1 and 2. The mill is operated in open circuit just for convenience to minimize the computation because of limited time available for course participants. A closed circuit mill could of course be easily simulated by MODSIM or MicroSim simply by including a classifier in the flowsheet. A scavenger bank could also easily be added. The data for the Mount Lyell ore in a format suitable for input to MODSIM or MicroSim is given in tables 1, 2 and 3. It is suggested that a reference flowrate of 100 t/hr be used with 3m³ cells in the rougher and 0,25 m³ cells in the cleaner bank. The Sutherland flotation model FLTN (standard in both MODSIM and MicroSim) should be selected and the liberation - enhanced Austin mill model GMIL must be used to simulate the liberation during milling. No other mill model will do this. Sutherland's flotation model requires the solid holdup to be specified and a value of 350 kg/m³ can be used for the purposes of this study.

Table 1. Particle size distribution of flotation plant feed.

SIZE microns	MASS %	% Cu
+425	2,126	1,00
-425 +300	0,768	0,43
-300 +212	5,144	0,45
-212 +150	12,807	0,90
-150 +106	16,592	1,20
-106 +75	10,490	1,67
-75 +53	9,151	2,03
-53 +38	5,937	8,46
-38 +32	6,822	5,31
-32 +24	2,141	2,53
-24 +17	5,398	2,13
-17 +12	5,312	2,06
-12 +8,6	2,964	
-8,6 +4,7	9,606	
-4,7	10,741	
	100,000	2,01

Table 2. Distribution of particle types as measured by QEM*SEM analysis.

SIZE	PARTICLE TYPE SPECIFIED BY % CHALCOPYRITE					
microns	0%	0 - 30%	30 - 60%	60 - 90%	90 - 100%	
+150	96,84	0,61	0,40	0,32	1,83	100
-150 +106	95,45	0,59	0,62	0,37	2,97	100
-106 +75	94,26	0,30	0,54	0,53	4,37	100
-75 +53	93,38	0,19	0,39	0,34	5,70	100
-53 +38	73,26	0,35	0,86	1,23	24,30	100
-38 +32	83,34	0,25	0,35	0,48	15,58	100
-32 +24	92,12	0,08	0,15	0,20	7,45	100
-24 +17	93,40	0,03	0,13	0,10	6,34	100
-17 +12	93,65	0,03	0,09	0,06	6,17	100
-12	93,83	0,03	0,11	0,05	5,98	100

Table 3. Distribution of particle types among flotation constants.

FLOTATION RATE CONSTANT	PARTICLE TYPE				
	0	0 -30%	30 - 60%	60 -90%	90 - 100%
Fast floating 6 min ⁻¹	0,0	0,51	0,67	0,83	0,94
Slow floating 0,032 min ⁻¹	0,1	0,49	0,33	0,17	0,06
Entraining 0,001min ⁻¹	0,9	0,0	0,0	0,0	0,0
	1,0	1,00	1,00	1,00	1,00

An interesting simple exercise is the examination of the effect of varying the ore texture parameter Φ over the range 10 - 100 and also the residence time (size) of the regrind mill. The associated grade recovery curves can easily be generated.

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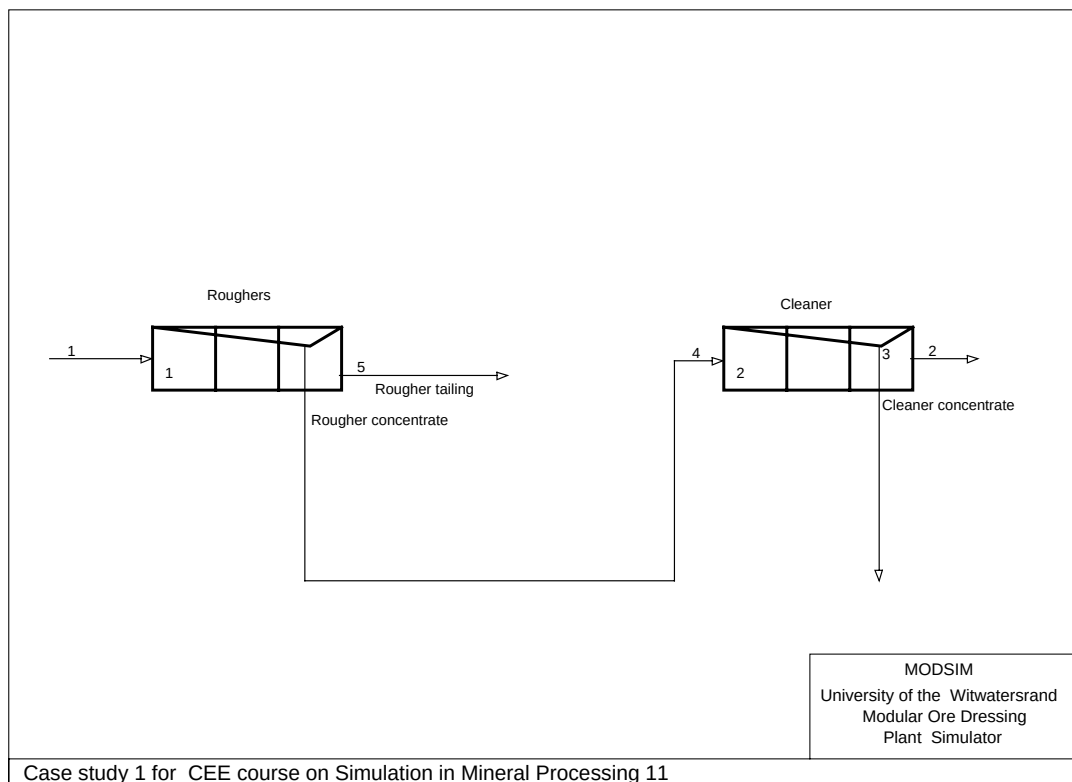


Figure 4.3 A simple rougher cleaner circuit without regrind.

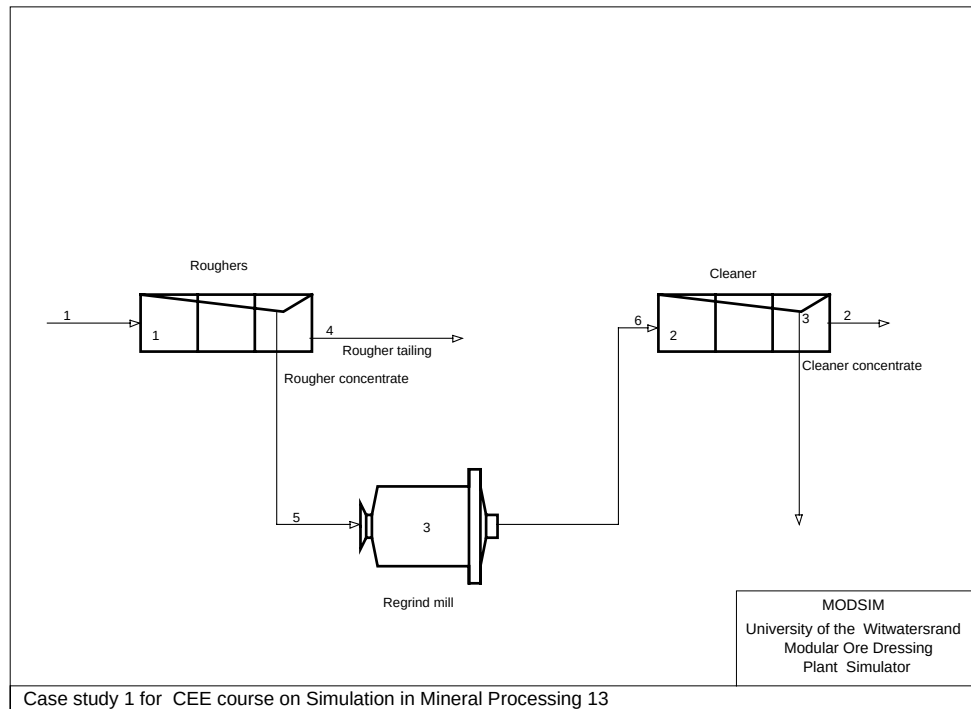


Figure 4.4 A regrind mill inserted into the circuit of Figure 1 should liberate locked mineral in the rougher concentrate.

EXAMPLE 6 Optimization of an existing crushing and screening circuit

The paper entitled "Simulation - the modern cost-effective way to solve crusher circuit processing problems" by R P King (Int Jnl of Mineral Processing 29(1990)249-265 discusses the use of simulation to solve plant processing limitations. Investigate the flowsheet show in the paper and assess the validity of the conclusions that are drawn in the paper.

The plant processes 2000 tons/hr of coarse ore reclaimed from the primary crusher product stockpile. The particle size distribution in the feed is given in Table 1.

TABLE 1. SIZE DISTRIBUTION IN PLANT FEED

MESH SIZE mm	% PASSING
237	98,4
168	89
119	75
83,8	64
59,3	54
41,9	45
27,6	38,5
21,0	33
14,8	28,5
10,5	25
7,41	22,5
5,25	20,5
3,71	19,2
2,62	18

The equipment operating parameters are summarized in Table 2.

An intensive sampling campaign on this plant has confirmed that MODSIM or MicroSim can be used to simulate its operation accurately in that the simulations match the measured size distributions and mass flowrates in the internal plant streams

Use one of the simulators to simulate the existing operation and then explore possibilities to improve the plant performance in a cost effective manner.

A new operating target is 80% passing 7 mm in the final product and an increase in tonnage to 2500 ton/hr. The work index for the ore is 9 kWhr/t.

Table 2 IMPORTANT OPERATING VARIABLES IN THE EXISTING PLANT

Unit	Type	Operating parameters
Secondary crushers	7ft standard Symons cone crushers	Closed-side settings: West 49.1mm East 51.7mm Installed power: West 171kW East 158kW
Tertiary screens	Double-deck polyurethane	Top deck: 49mm square opening 41mm ribs 23.4% open area 5.486 m x 2.134 m 18° inclination Lower deck: 15.2mm square opening 10.3 mmribs 25.01% open area 6.094 m x 2.134 m 18° inclination
Tertiary crusher	7 ft short head cone crushers	Closed-side settings: West: No.1 13.1mm No.2 11.9mm East: No.3 13.7mm No.4 12.6mm Installed power: West 151kW East 145kW
Quaternary crushers	7 ft short-head cone crushers	Closed-side settings: No.1 8.17mm No.2 8.6mm Installed power: No1 158kW No2 148kW
Quaternary screens	6 x single-deck polyurethane	20mm square opening 11.5mm ribs 27.7% open area 5.48m x 2.0m 18° inclination

EXAMPLE 7 Screening operations

Calculate the size distribution in the overflow and underflow streams from the following screening operation..

Data: Feed tonnage 293 tons/hr
 Screen length 5.48 m
 Screen width 2.08 m
 Mesh aperture 2.00 cm
 Wire diameter 1.15 cm
 Screen angle 18°
 Bulk density of feed material 1600 kg/m^3

Particle size distribution in the feed

Size cms	% passing
4.19	100.0
2.96	98.2
2.10	89.9
1.48	71.6
1.05	52.6
0.741	38.2
0.524	28.7
0.371	22.4
0.262	18.1

Calculate the actual transmission efficiency and the area utilization factor for this screen.

EXAMPLE 8: Screen selection

Select a suitable screen to handle the following duty.

265 metric tons of iron ore which has specific gravity 3.47, bulk density 2082 kg/m^3 and carries 8% moisture. The required separation size is 12.5 mm. The size analysis of the material is

Mesh size mm	Cumulative % passing
38	100
25	98
19	92
12.5	65
6.3	33

The old fashioned manual calculation procedure for this typical design problem is described in the SME handbook on pages 3E-9 and 3E-13. You will find it easier and more accurate to make the selection using MODSIM. To do so the recommended procedure would be to use the simple screen model SCRn which is based on the traditional procedure to establish a preliminary design and then to simulate the proposed design using the more accurate and more reliable Karra model to fine tune the design.

EXAMPLE 9: Standard crusher circuits

The attached pages are taken from the latest Nordberg product brochure. Four different crusher flowsheets are shown. Simulate the performance of each of these circuits. Calculate the size distribution in the final product from each flowsheet and use the standard equipment capacities from the class notes to assess whether the equipment sizes that are shown are appropriate. Calculate the energy required for the crushing operations if the crusher work index of the material is 12.5 kWhr/tonne in each case. Make what ever assumptions you consider appropriate if essential data items are missing.